

POPULATION PROPERTIES OF ECCENTRIC BINARY BLACK HOLES

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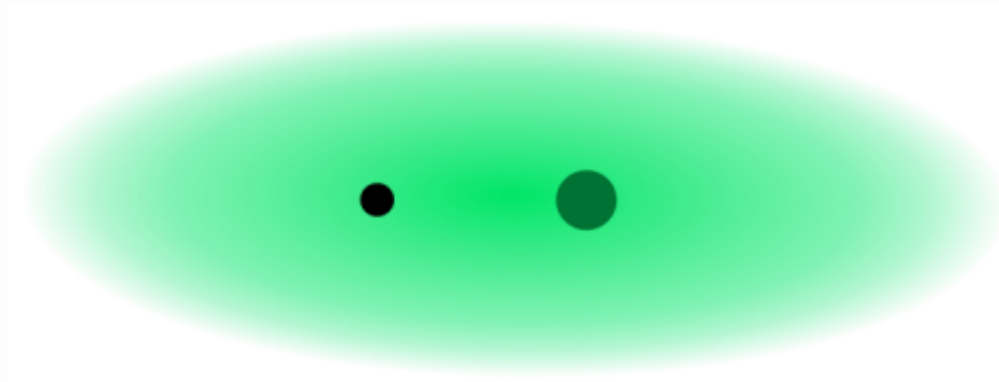
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WHY ARE WE INTERESTED IN ECCENTRIC BINARIES?

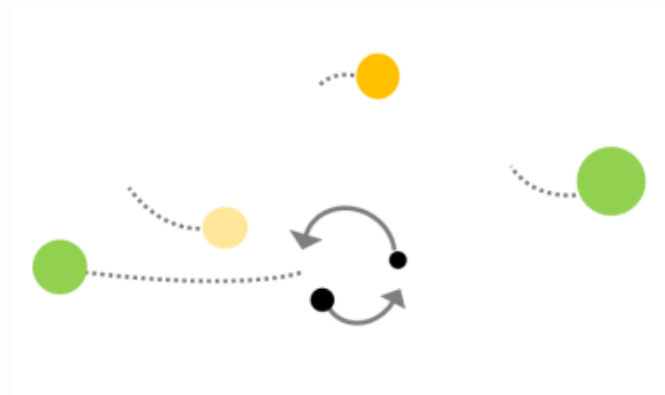
Predictions from N-body or Monte Carlo codes show that a **fraction of dynamically formed binaries are eccentric**

Common envelope



Dynamical formation in cluster

vs



Mandel & Farmer 2018

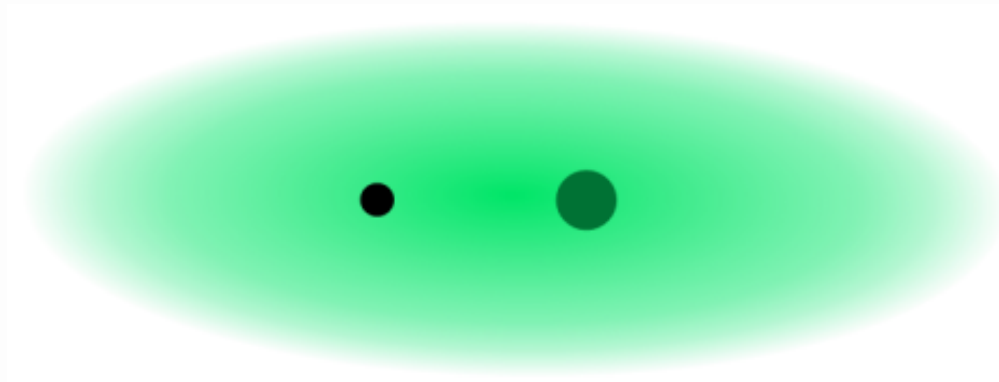
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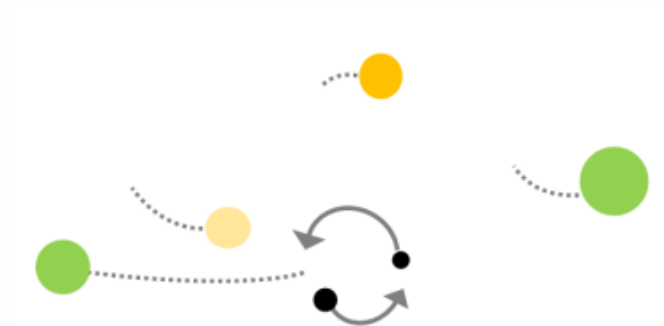
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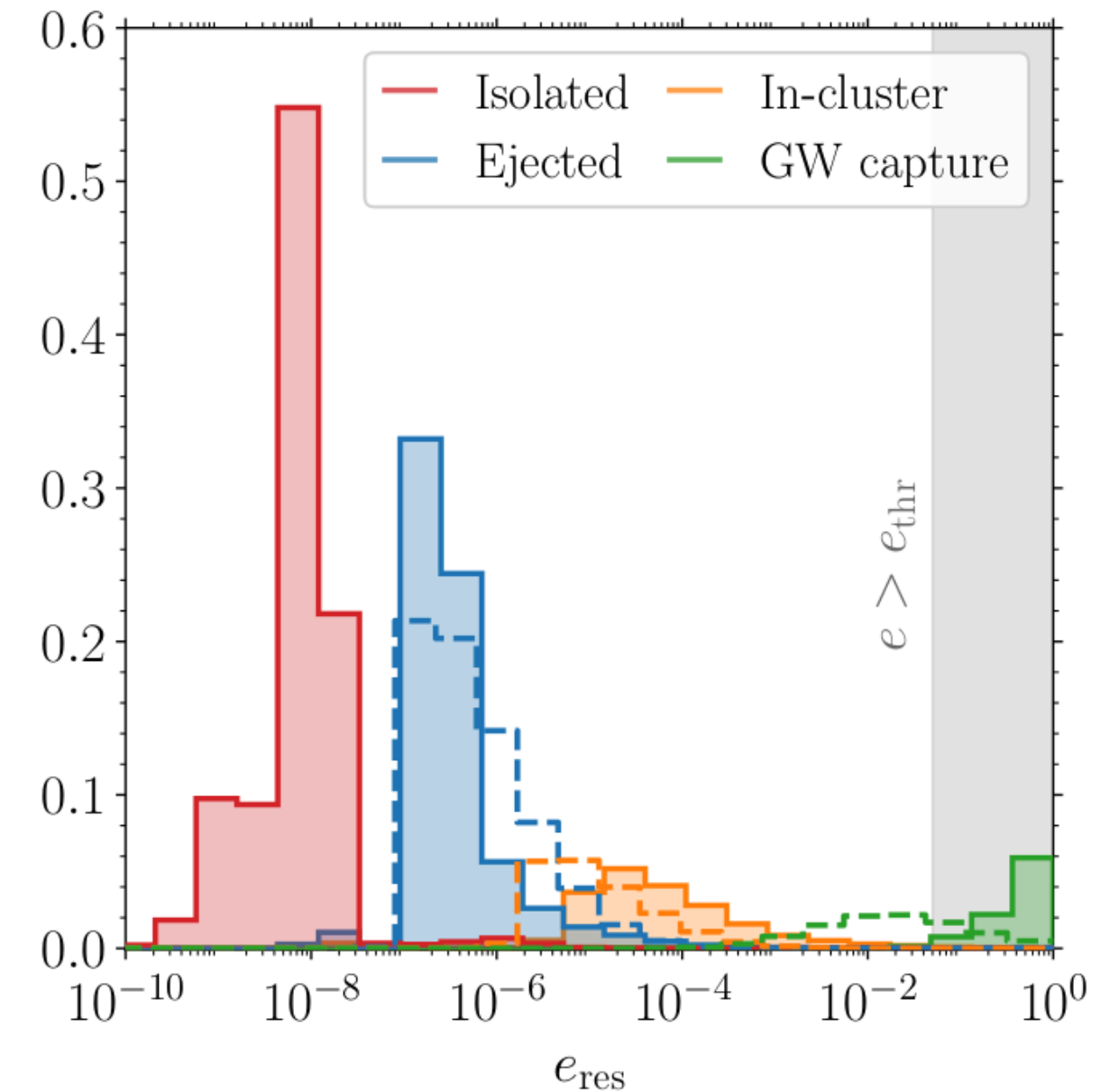
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Mandel & Farmer 2018

The precise origin of stellar mass BBHs is still unclear

But, eccentric GWs can **provide astronomers a unique window into the origins of stellar mass BBHs.**



Fumagalli+ 2024, Olejak+ 2020,
Kremer+ 2022, Kritos+ 2022

ACCELERATE INFERENCE USING MACHINE-LEARNING (DINGO)

Eccentric inference takes $O(\text{week})$ with
SEOBNRv4EHM and $O(\text{days})$ with SEOBNRv5EHM
(Ramos-Buades+ 2023, Gamboa+ 2024)

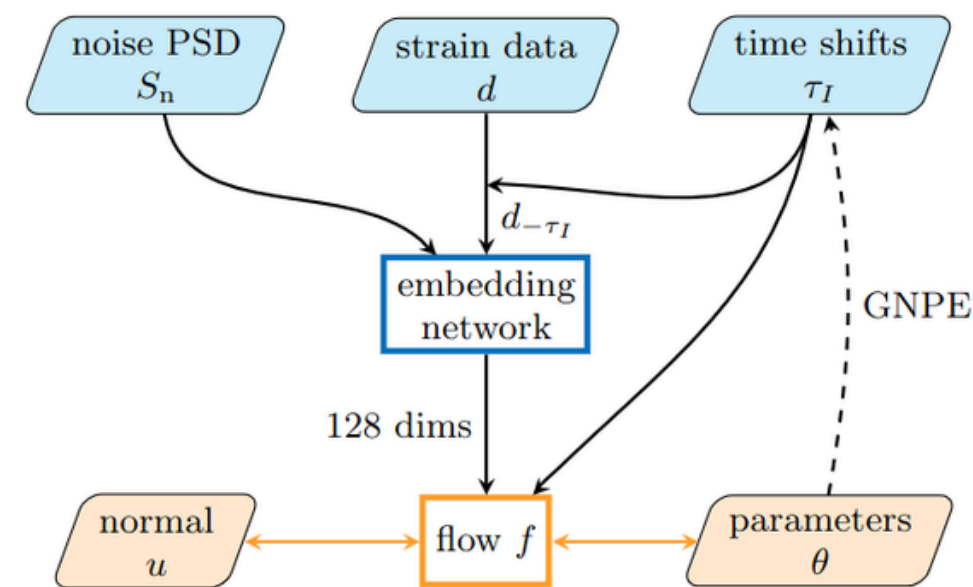
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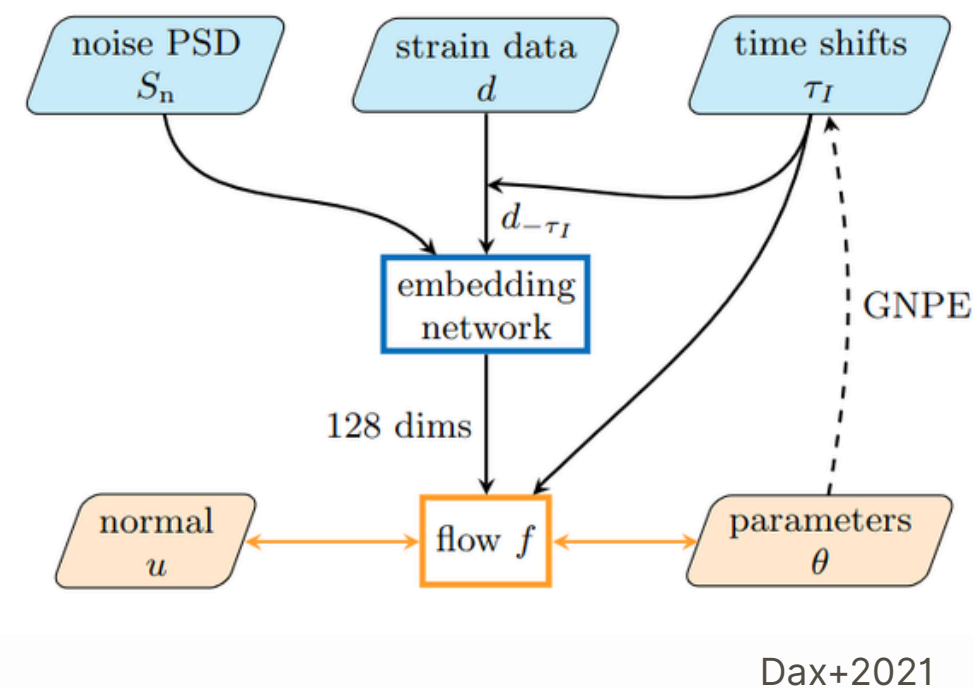
Dax+2021

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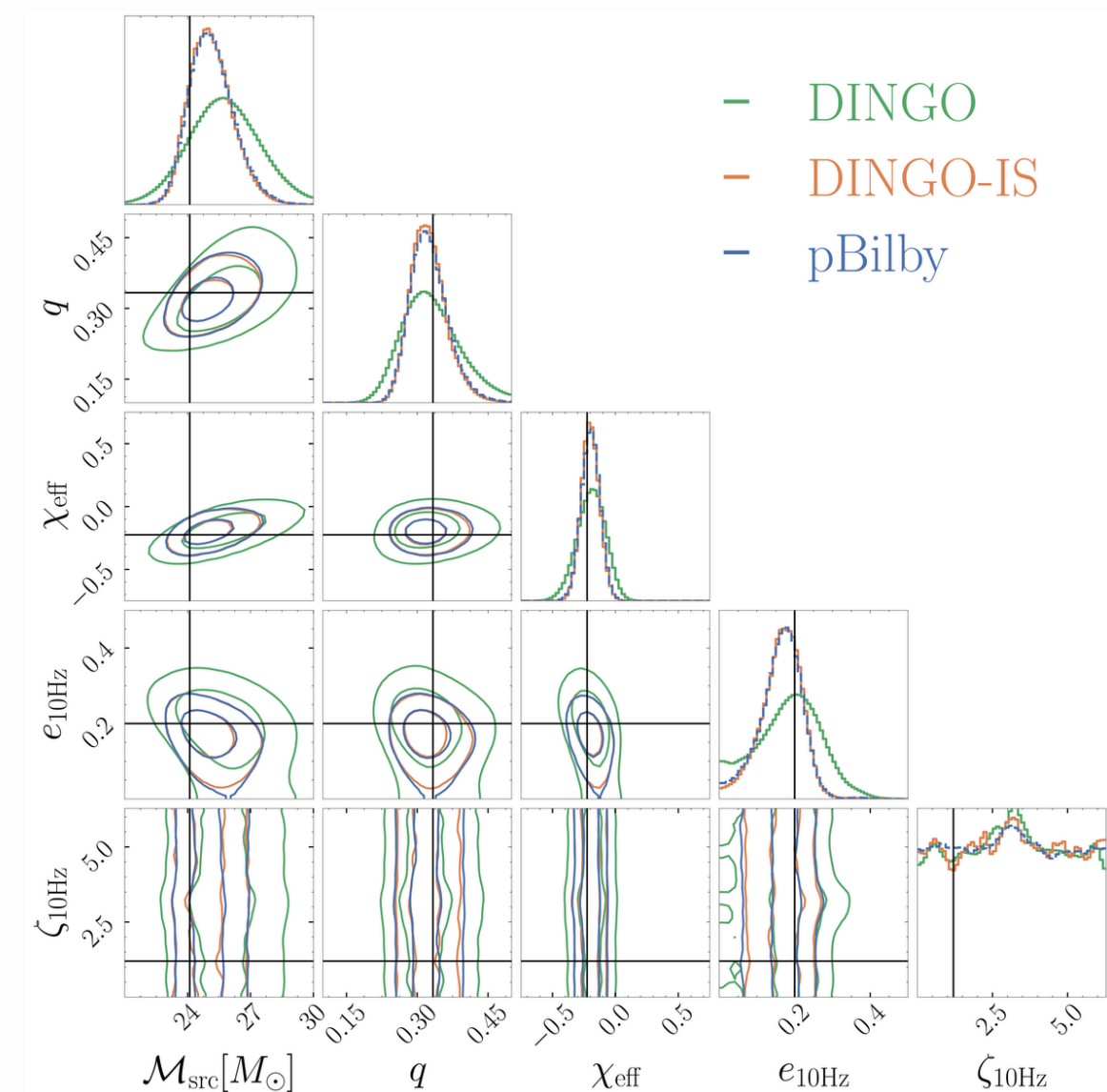
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2). **Importance sample** the proposal distribution to ensure accuracy with traditional methods
 $O(\text{hour})$

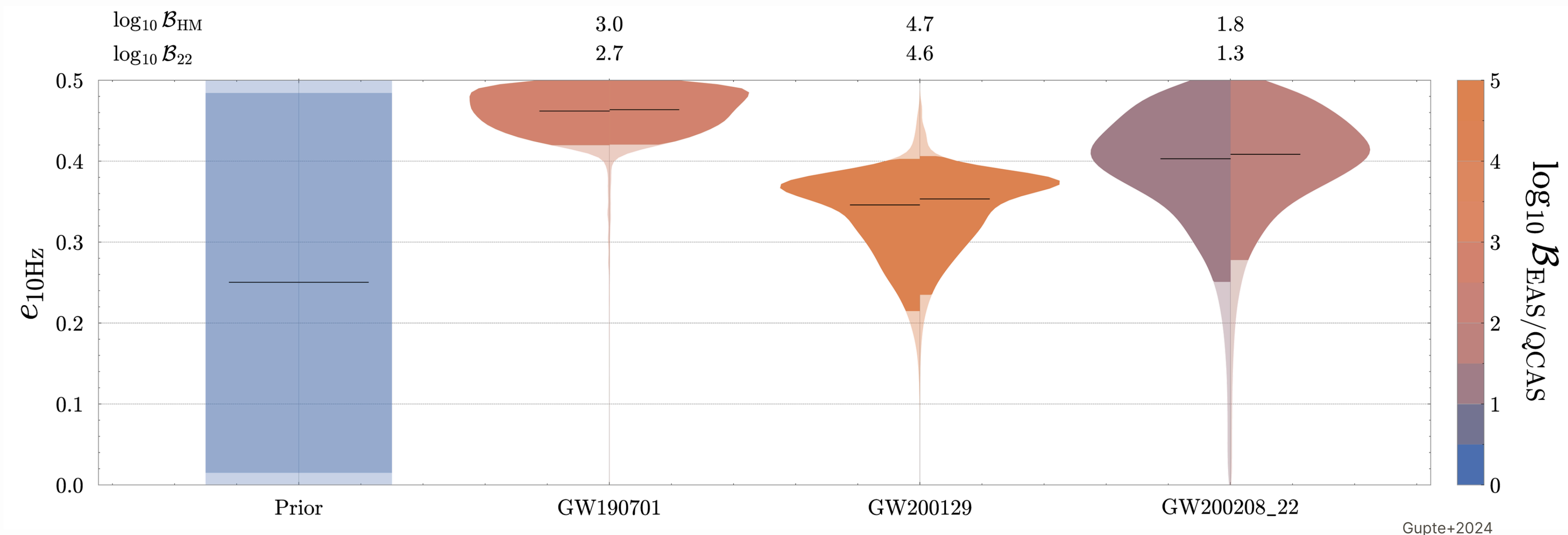


Dax+2022, Gupte+2024, Smith+ 2019

SO DO WE FIND ANYTHING INTERESTING? YES!

After analyzing 57 events from O3, we find that 3 of them have support for eccentricity. There have been other eccentric PE studies: Romero-Shaw 2019, 2020, 2022, Iglesias+2022, Sun+2015, O'Shea+ 2023, Ramos-Buades+ 2024 Wagner+2024, Morras+ 2025, Planas+ 2025

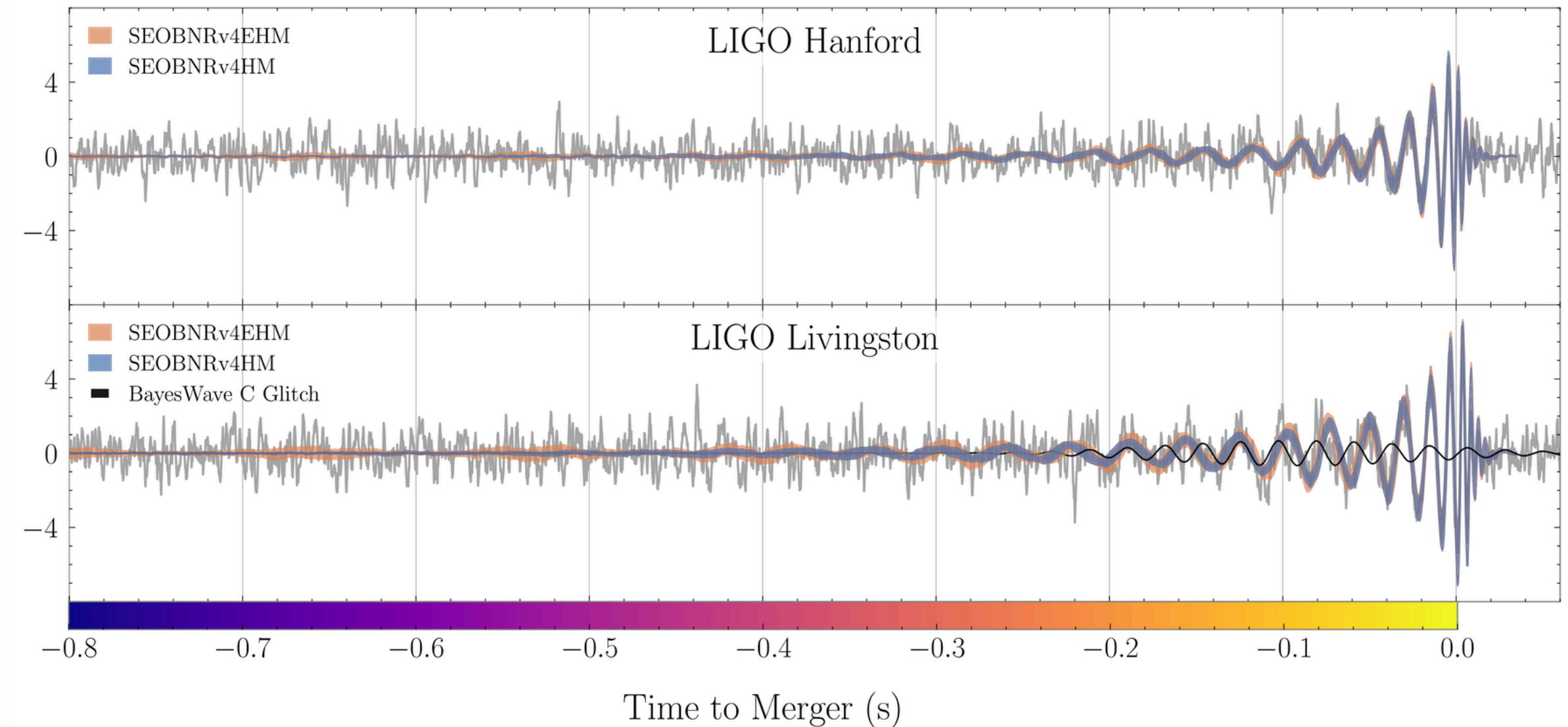
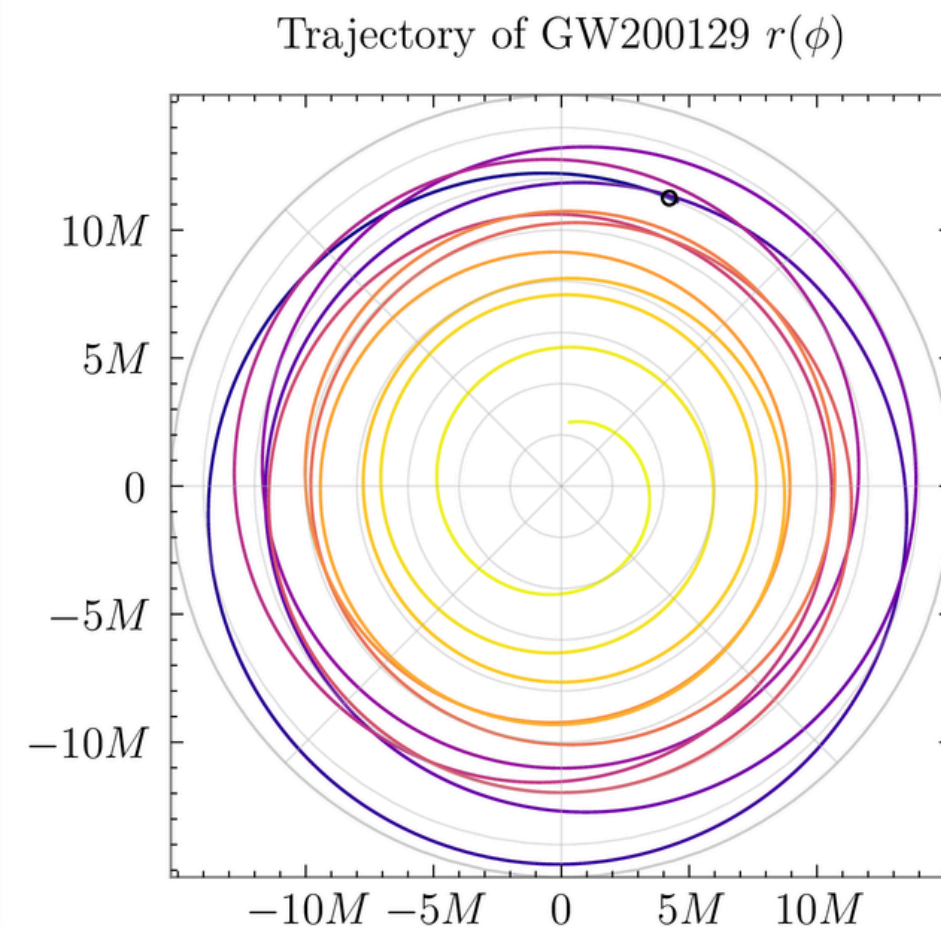
Our main contribution is to analyze a lot of events with a waveform model that includes two eccentric parameters. The new events in this work are GW200129 and GW190701.



A SPECIAL EVENT: GW200129

This event has a Bayes factor $>50,000$ between the eccentric-aligned spin vs quasi-circular aligned-spin hypothesis!

The detector projection (right) and dynamics (below) of GW200129. There are differences in the inspiral where the periastron passages lead to excess power.

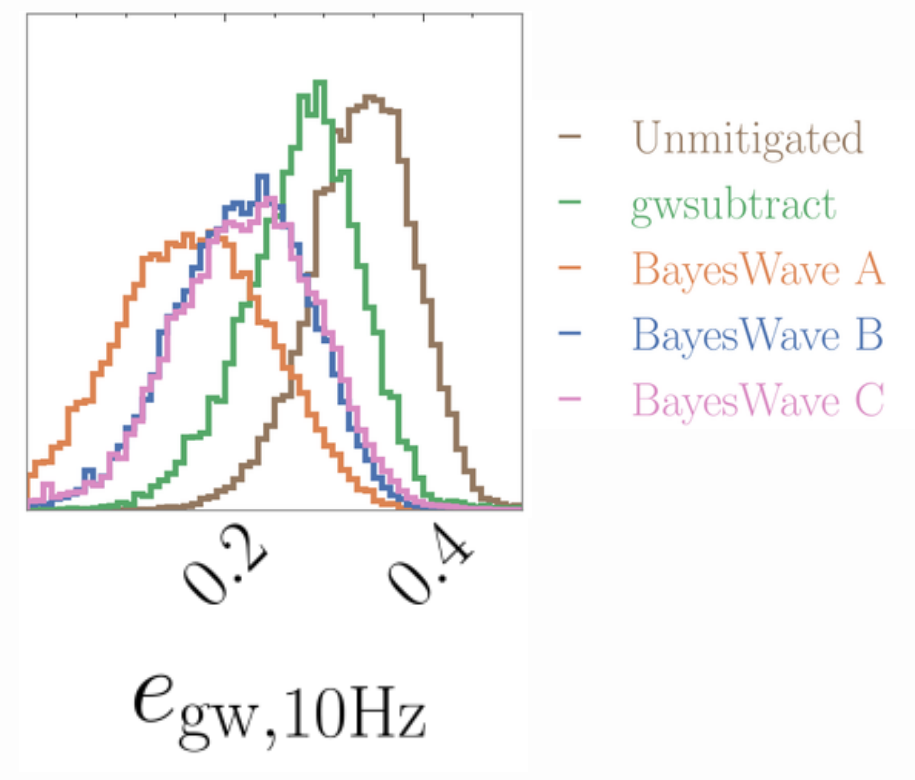


Gupte+2024

Given such a high Bayes factor, this definitely seems like an eccentric event. **But there are some important caveats....**

CAVEAT AND FEATURES OF GW200129

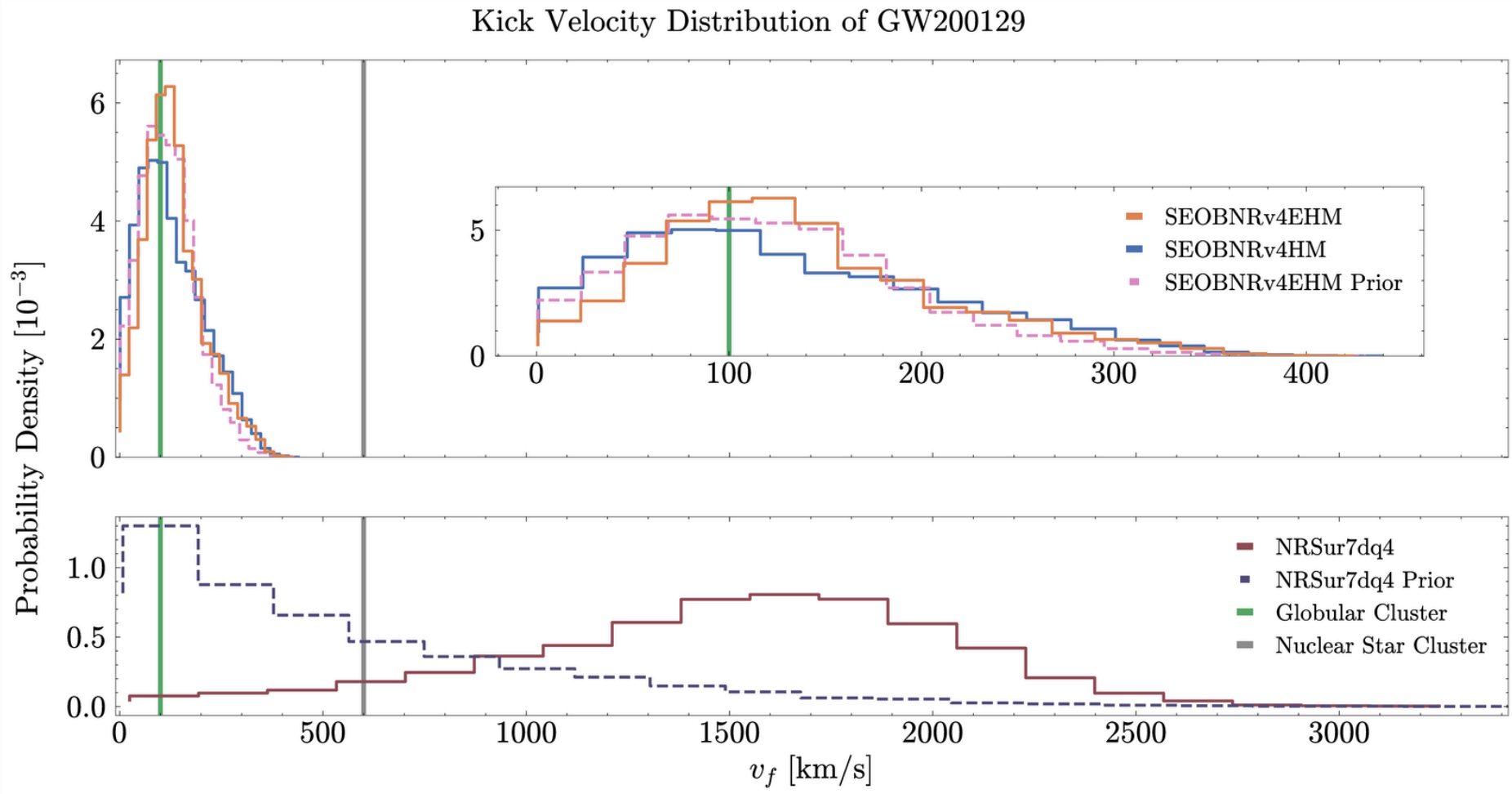
The level of eccentricity changes depending on how one subtracts the glitch (Davis+2022, Payne+2022).



NRSur7dq4 $\log_{10} \mathcal{B}$		
gwsubtract	Uniform	4.0
BayesWave A	Uniform	1.53
BayesWave B	Uniform	2.35
BayesWave C	Uniform	1.43
gwsubtract	Log-Uniform	3.23
BayesWave A	Log-Uniform	0.94
BayesWave B	Log-Uniform	1.70
BayesWave C	Log-Uniform	0.76

We compare to the quasi-circular spin-precessing hypothesis (Hanam+ 2021, Romero-Shaw+ Bustillo+ 2022))

Varma+ 2022 shows a kick velocity in GW200129. However, under the eccentric aligned-spin hypothesis there is no longer an appreciable kick.



This does **NOT** mean there isn't a kick. We need an eccentric spin-precessing model to determine this

HOW DO WE ASSESS THE SIGNIFICANCE OF THE EVENT?

The Bayes factors are not the end of the story. Using a uniform prior is ok, but we know from astrophysics that non-eccentric channels *should* contribute more mergers.

$$\mathcal{O}_{\text{E/QC}} = \frac{R_{\text{E}}}{R_{\text{QC}}} \frac{p(d|\mathcal{H}_{\text{E}})}{p(d|\mathcal{H}_{\text{QC}})} \longrightarrow \tilde{p}_{\text{ecc}} = \frac{\mathcal{O}_{\text{E/QC}}}{1 + \mathcal{O}_{\text{E/QC}}}$$

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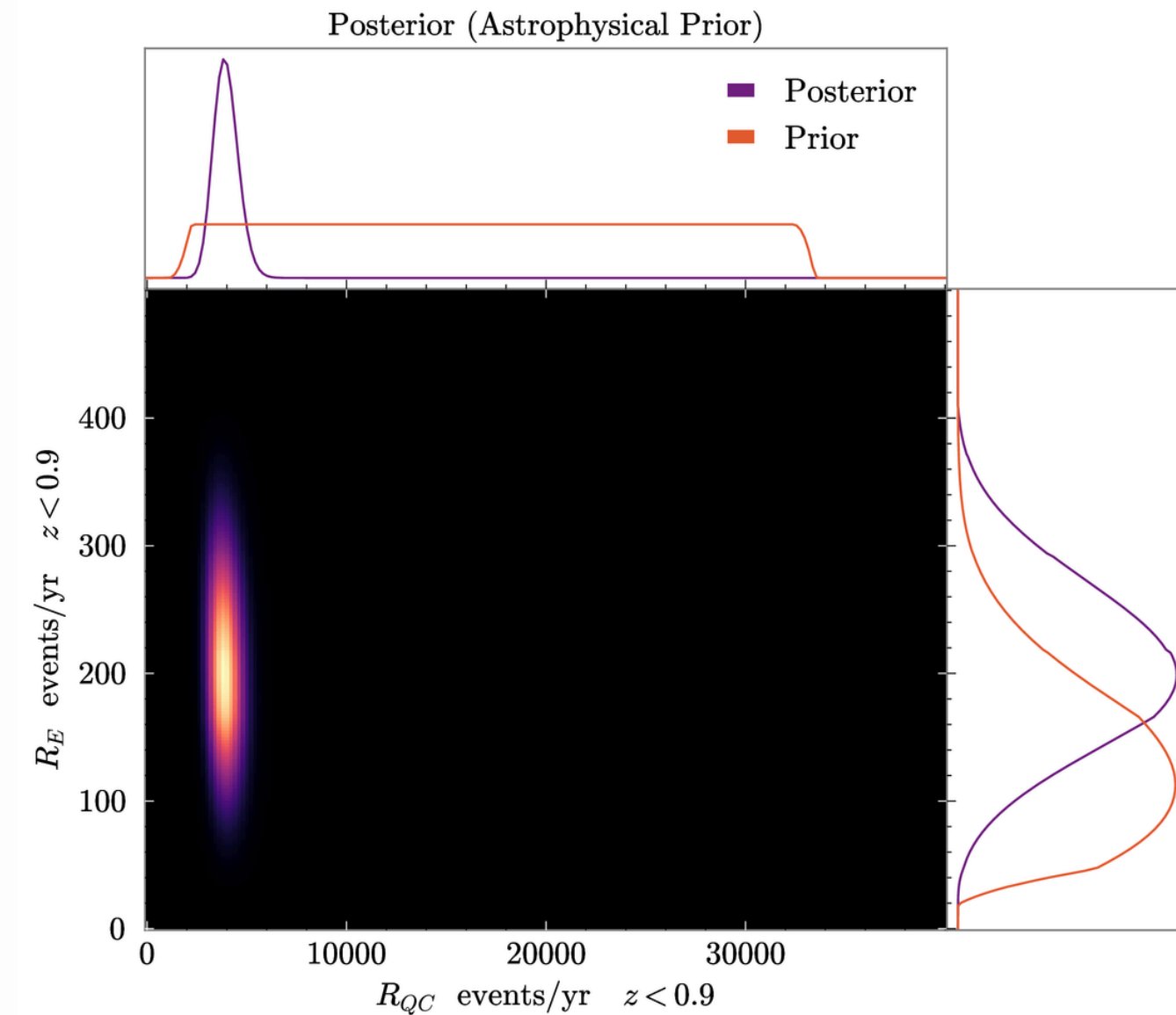
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The odds ratio isn't the end of the story either! What if we instead update our prior belief on the rates using GW observations? We can then marginalize over this rate (Farr+ 2013).

$$p(R_E, R_{QC}, \{g_i\}|\{d_i\}) \longrightarrow p_{ecc} = p(g_m = 1|\{d_i\})$$

$$p_{ecc} = 0.75\text{--}0.9995 \text{ for GW200129, } 0.97 \text{ for GW190701, } 0.72 \text{ for GW200208_22}$$



We see that GWs help constrain the rates of quasi-circular events quite well. In addition, the rate of eccentric events is pushed upwards by the GW data.

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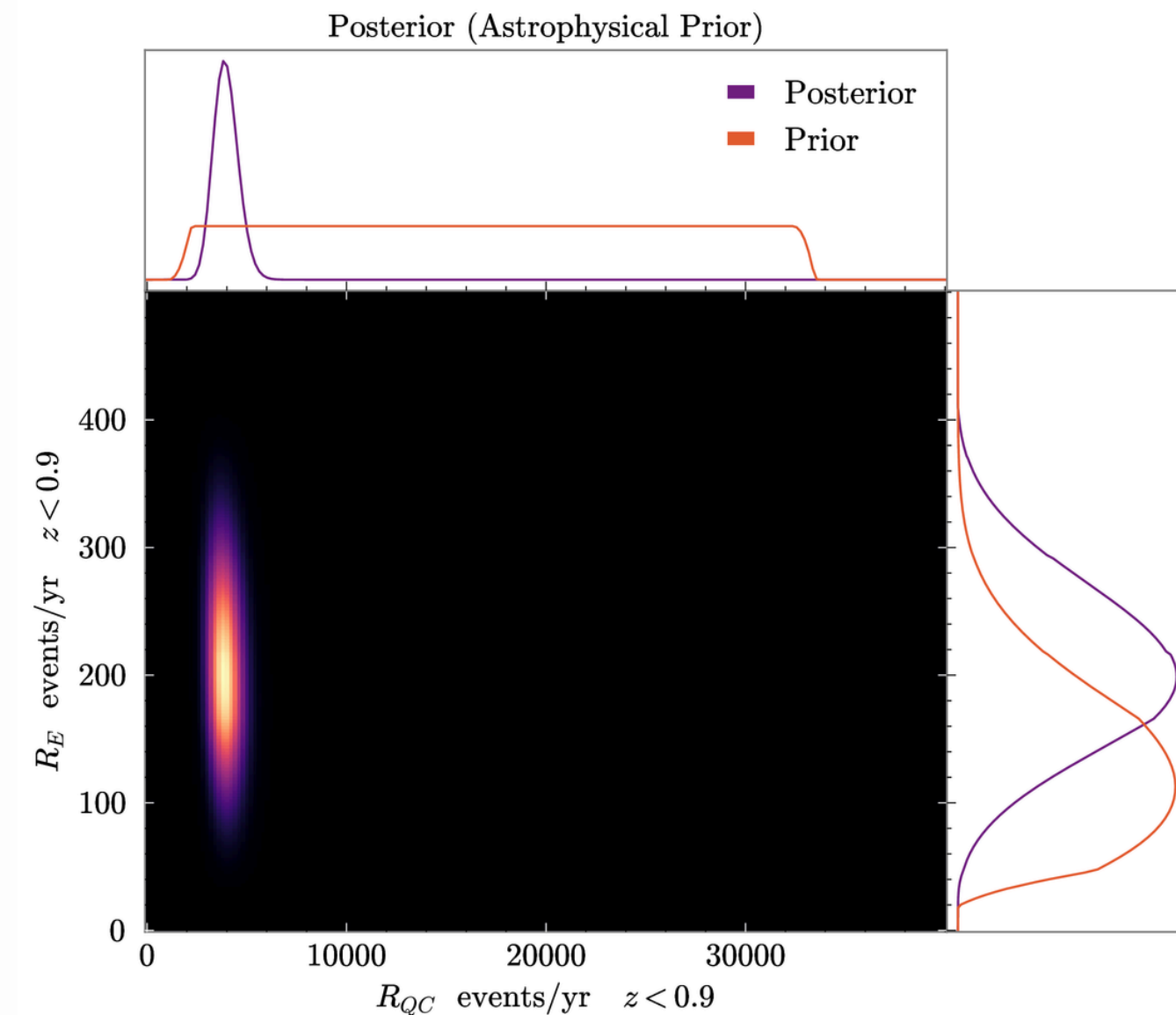
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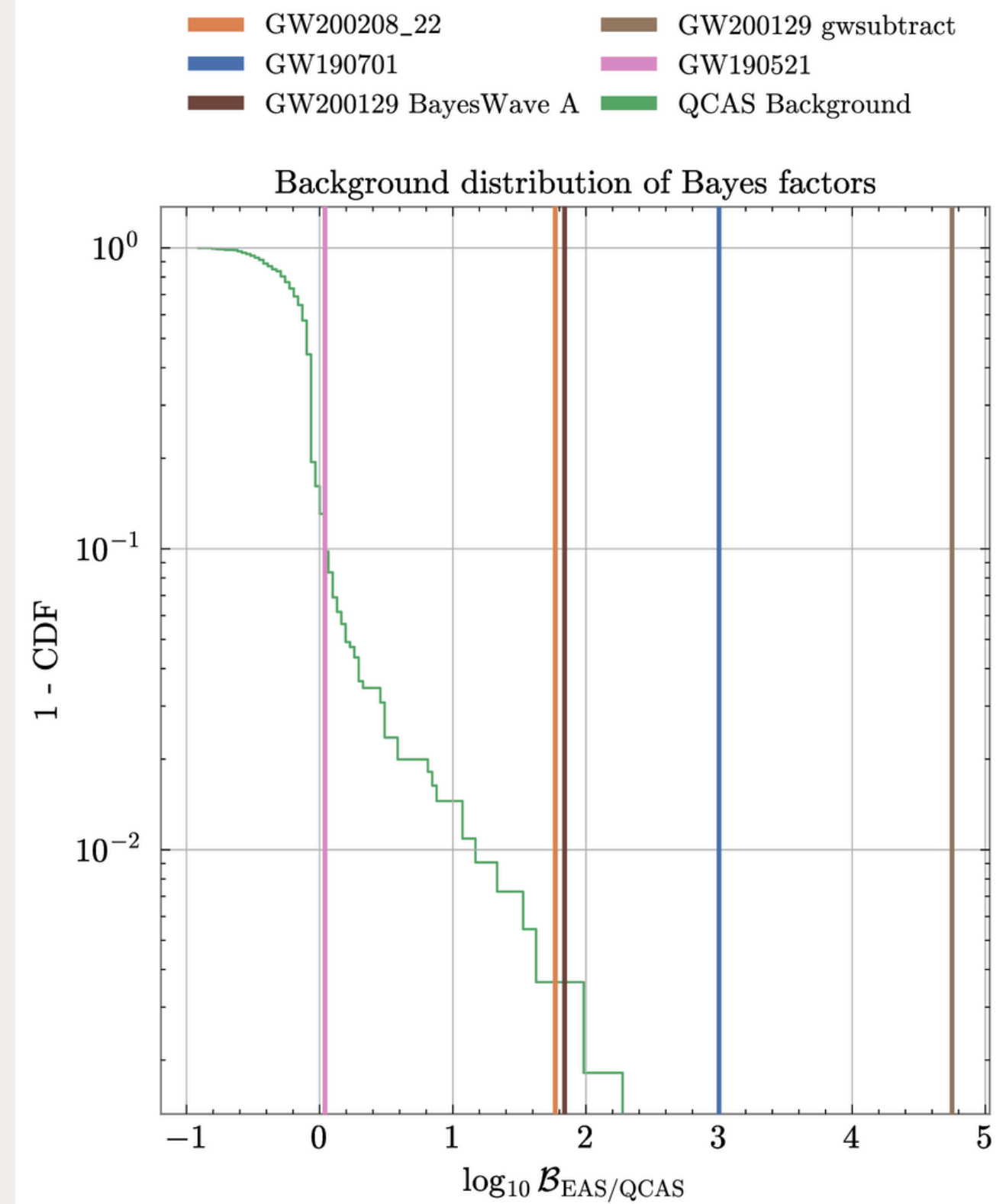
$$p_{ecc, \text{pop}} > 1 - 4.7 \times 10^{-3} \quad \text{or} \quad 1 - 8.4 \times 10^{-8}$$



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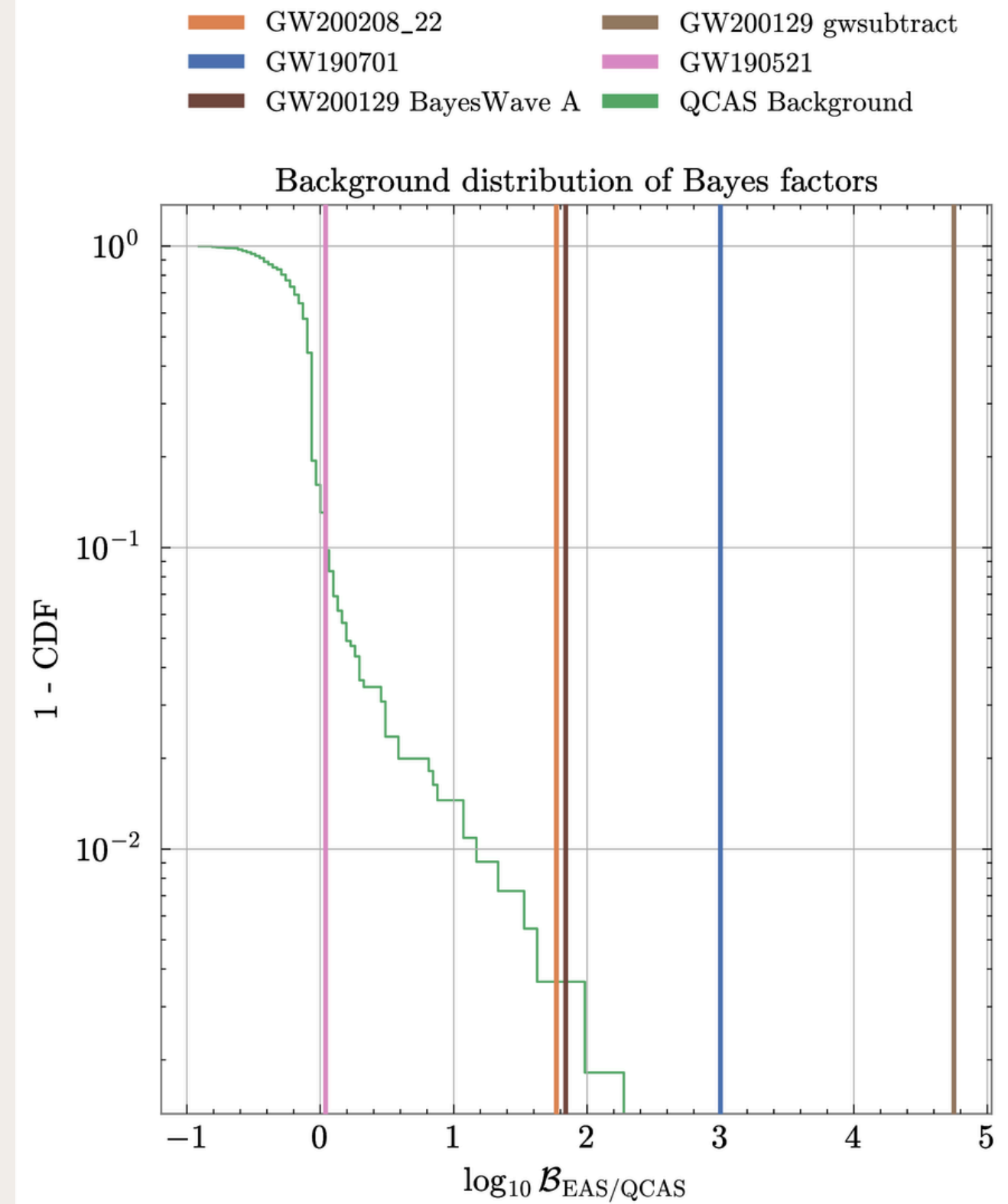
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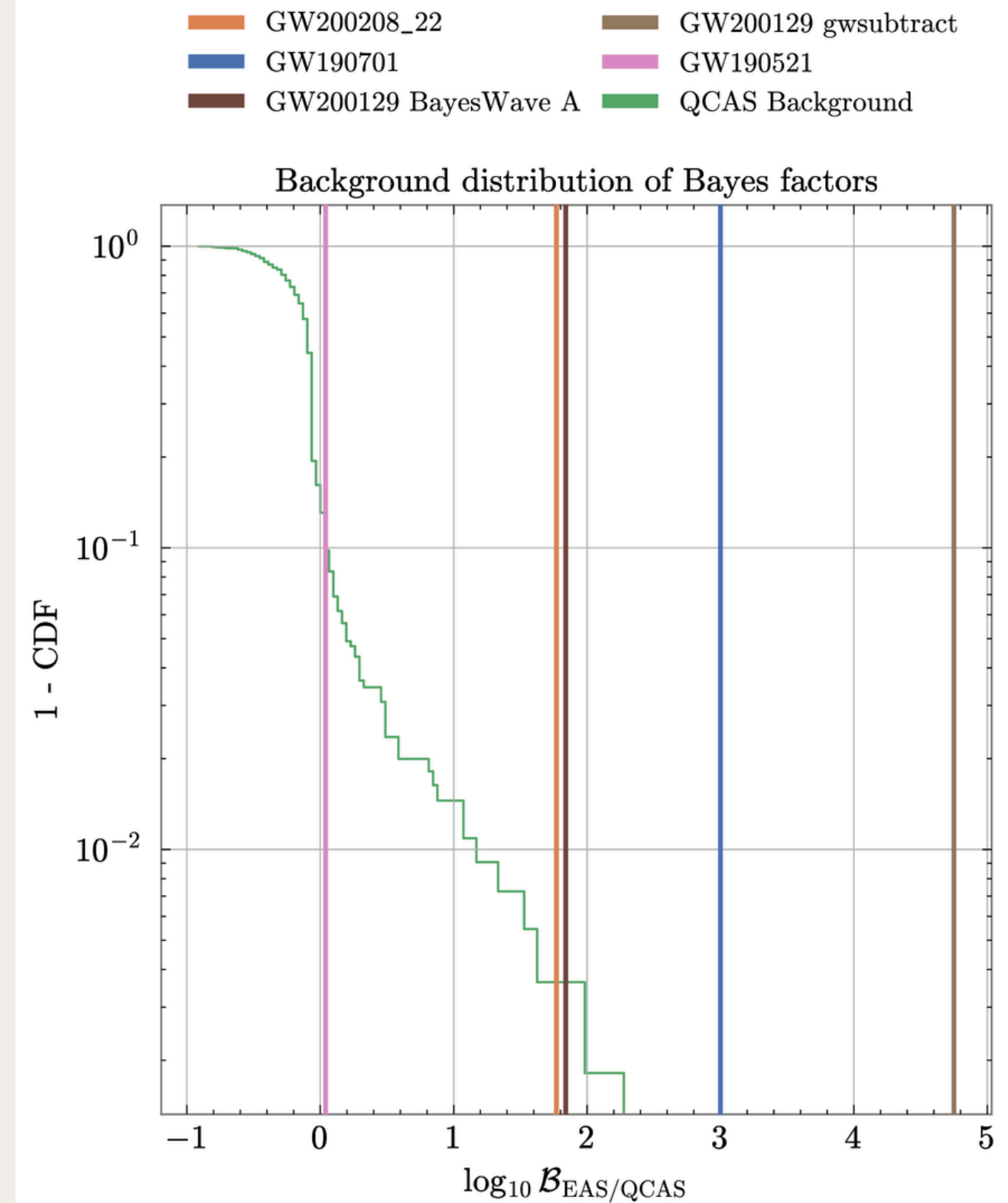
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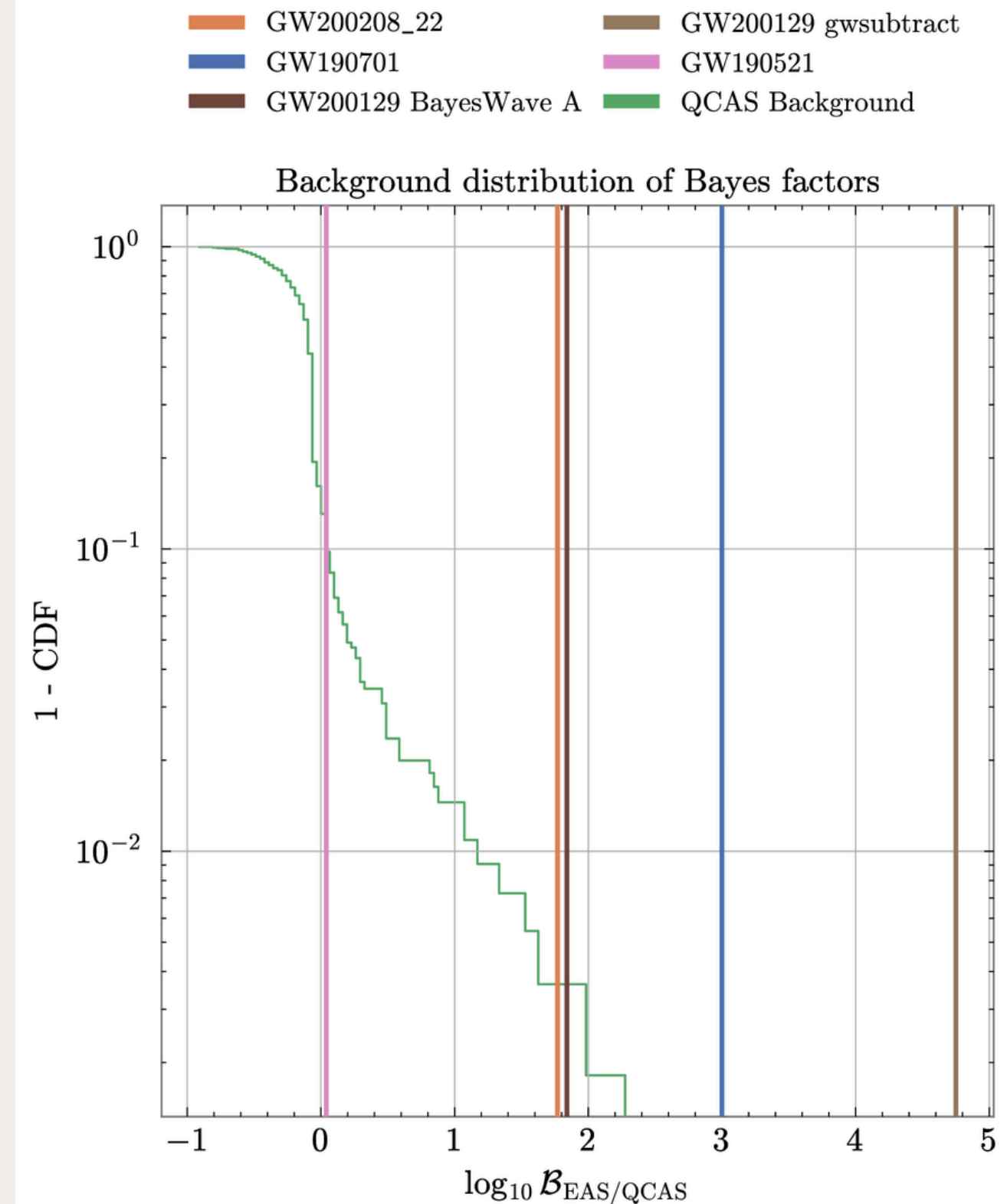
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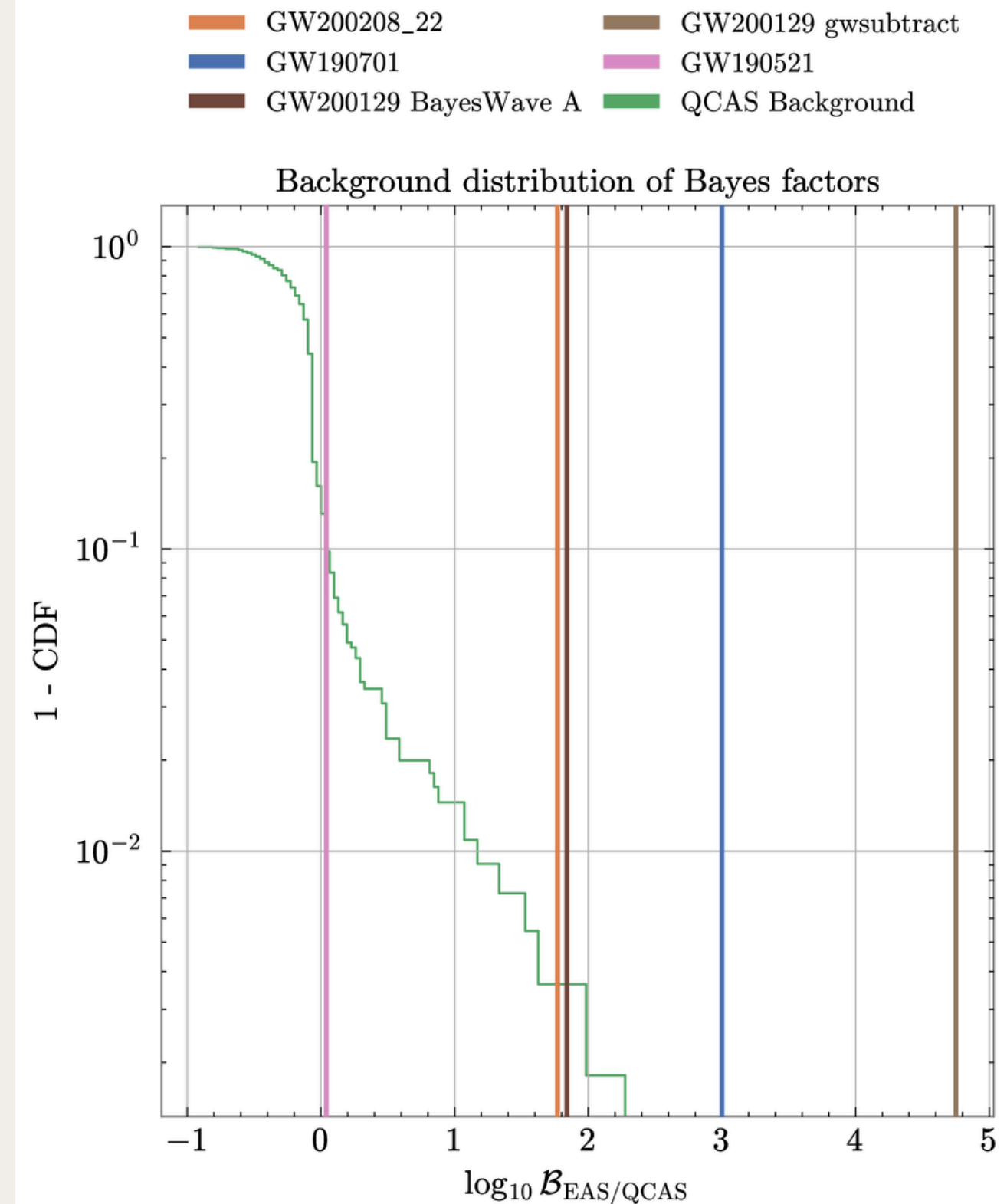
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- 05** We can extend this by injecting a spin-precessing distribution with real noise to compute a background distribution which is robust to systematics (still in progress). It can useful to evidence networks here (Jeffrey+ 2023)



A REVIEW OF GW POPULATION INFERENCE

We want the population posterior

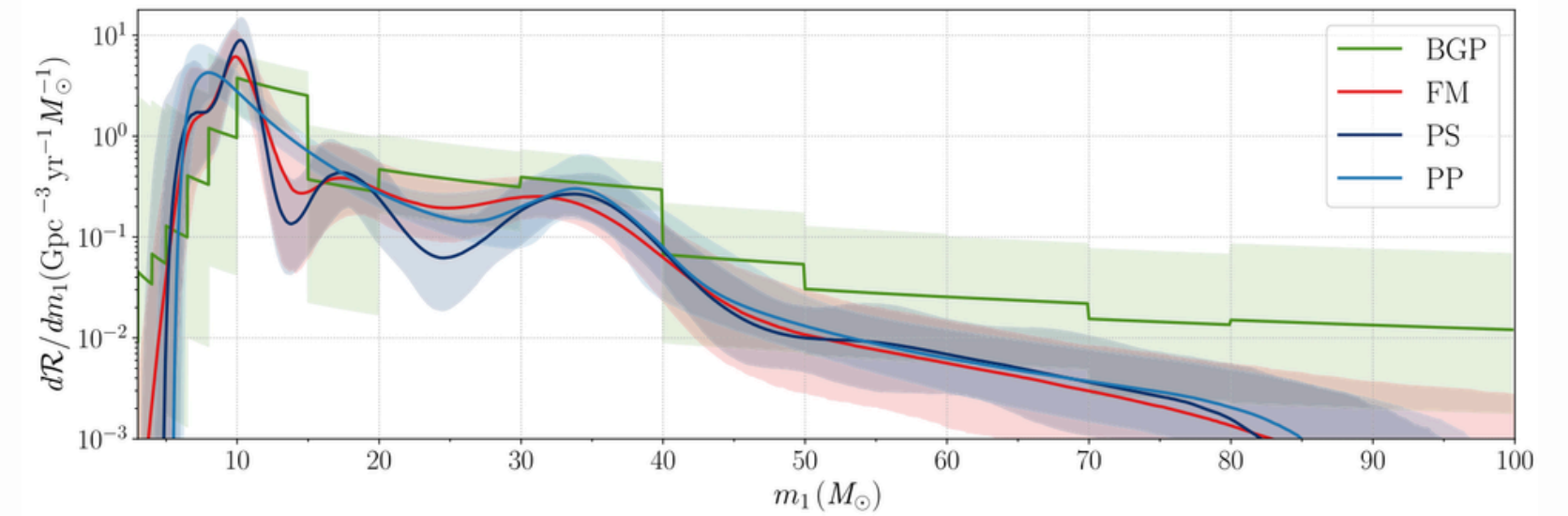
$$p(\lambda|\{d_i\}) = \frac{p(\{d_i\}|\lambda)p(\lambda)}{p(\{d_i\})}$$

The population likelihood is given by

$$p(\{d_i\}|\lambda) = \prod_i \frac{\int d\theta p(d|\theta)p(\theta|\lambda)}{\int d\theta p_{\text{det}}(\theta)p(\theta|\lambda)}$$

We can evaluate this using Monte Carlo

The numerator come from the posteriors of individual GW events and the denominator comes from a mock injection campaign.



LVK Collaboration 2022

How do we configure our population models?

We want to capture the relevant physical features of eccentric populations. We could also use a flexible search.

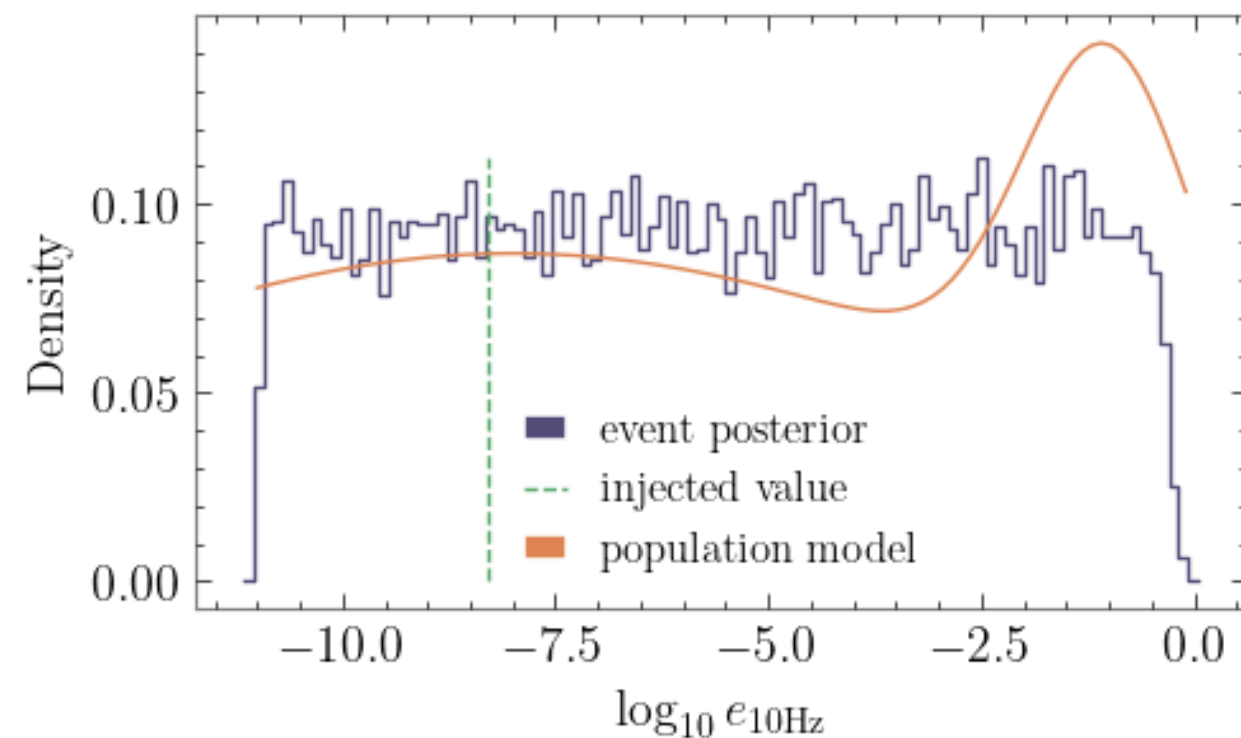
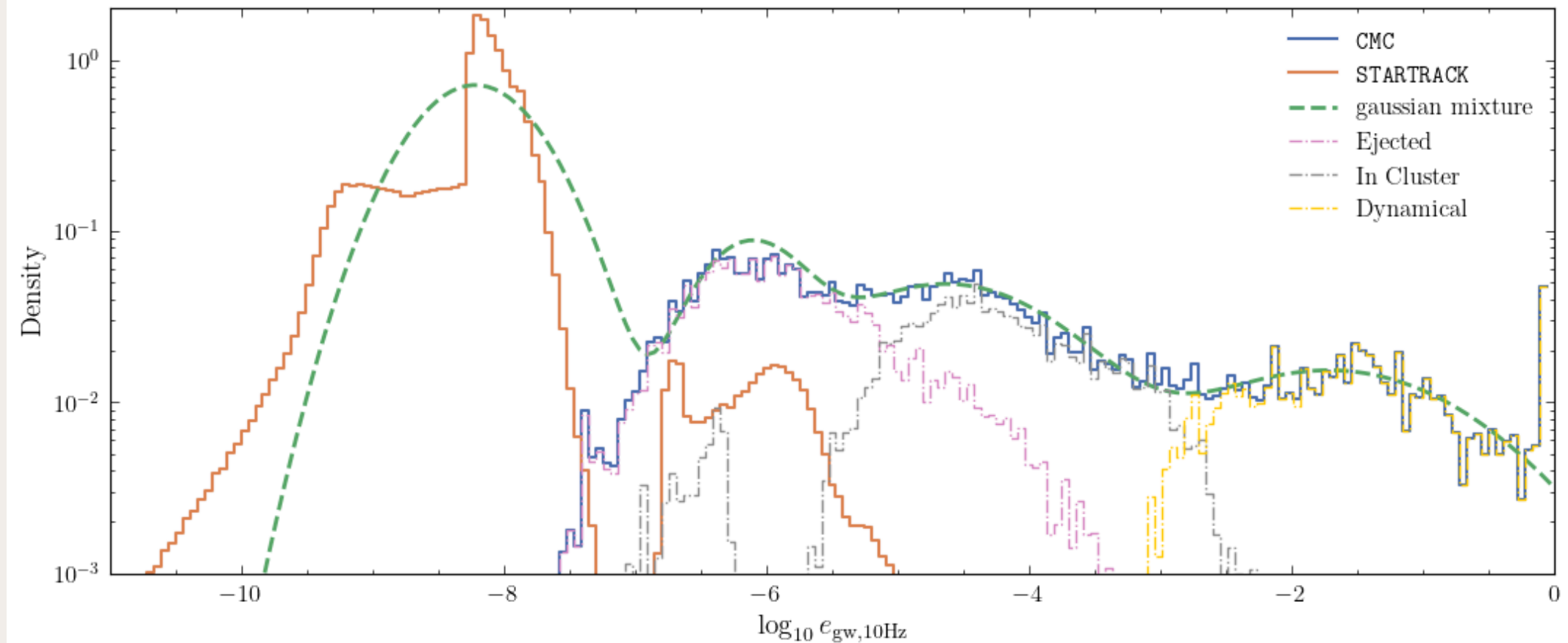
How do we compute the selection function?

Usually, the LVK computes the selection function by injecting signals and recovering with each pipeline. However, this can be expensive.

WHICH POPULATION MODELS ARE WE CONSIDERING?

One option is to use a [Gaussian mixture model](#) fitting the isolated binaries and dynamical binaries from STARTRACK (Belczynski+ 2001, Olejak+ 2022) and Cluster Monte Carlo (Rodriguez+ 2018, Kremer+ 2020)

$$p(e|\lambda) = (1 - \beta) \mathcal{N}(\log e | \mu_{\text{iso}}, \sigma_{\text{iso}}) + \beta \sum_{i \in \text{channels}} \mathcal{N}(\log e | \mu_i, \sigma_i)$$



Another option is to model only the eccentric component and place everything else into a separate population

$$p(e|\lambda) = (1 - \beta) \mathcal{G}(\log e | \mu_{\text{iso}}, \sigma_{\text{iso}}, \alpha_{\text{iso}}) + \beta \mathcal{N}(\log e | \mu_{\text{iso}}, \sigma_{\text{iso}})$$

In this presentation we will focus on the second for testing purposes.

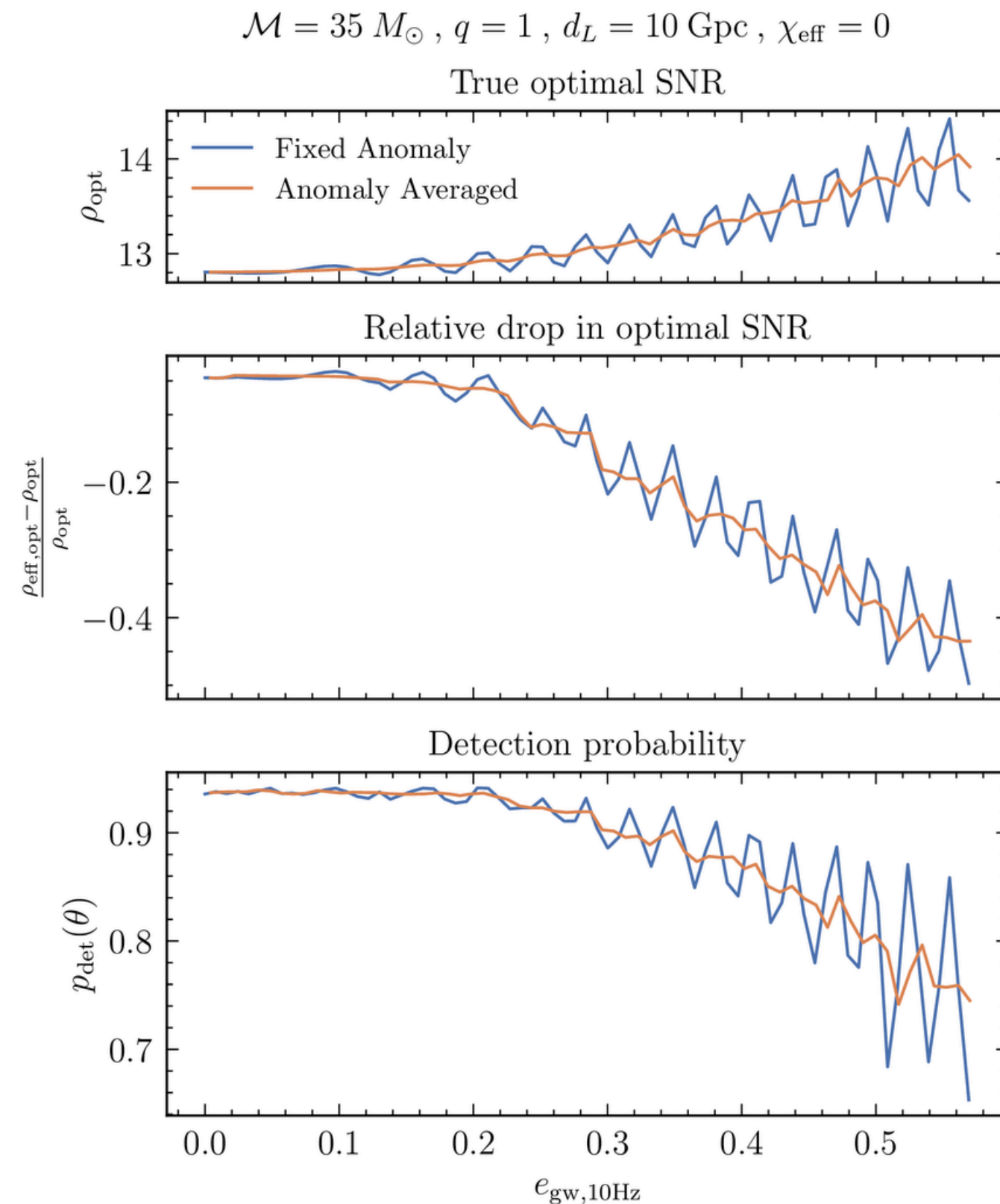
THE SELECTION FUNCTION

01 We miss a fraction of eccentric events since we don't use quasi-circular template banks

02 To quantify this we find the best fit optimal SNR from a pycbc quasi-circular template bank.

03 We assume the matched-filter SNR is distributed around this best fit SNR (Gerosa+2024)

$$p_{\text{det}}(\theta) = \int_{\rho_{\text{thresh}}}^{\infty} p(\rho_{\text{mf}}|\theta) d\rho_{\text{mf}}$$



Gupte+2025 (in prep)
Nee+2024

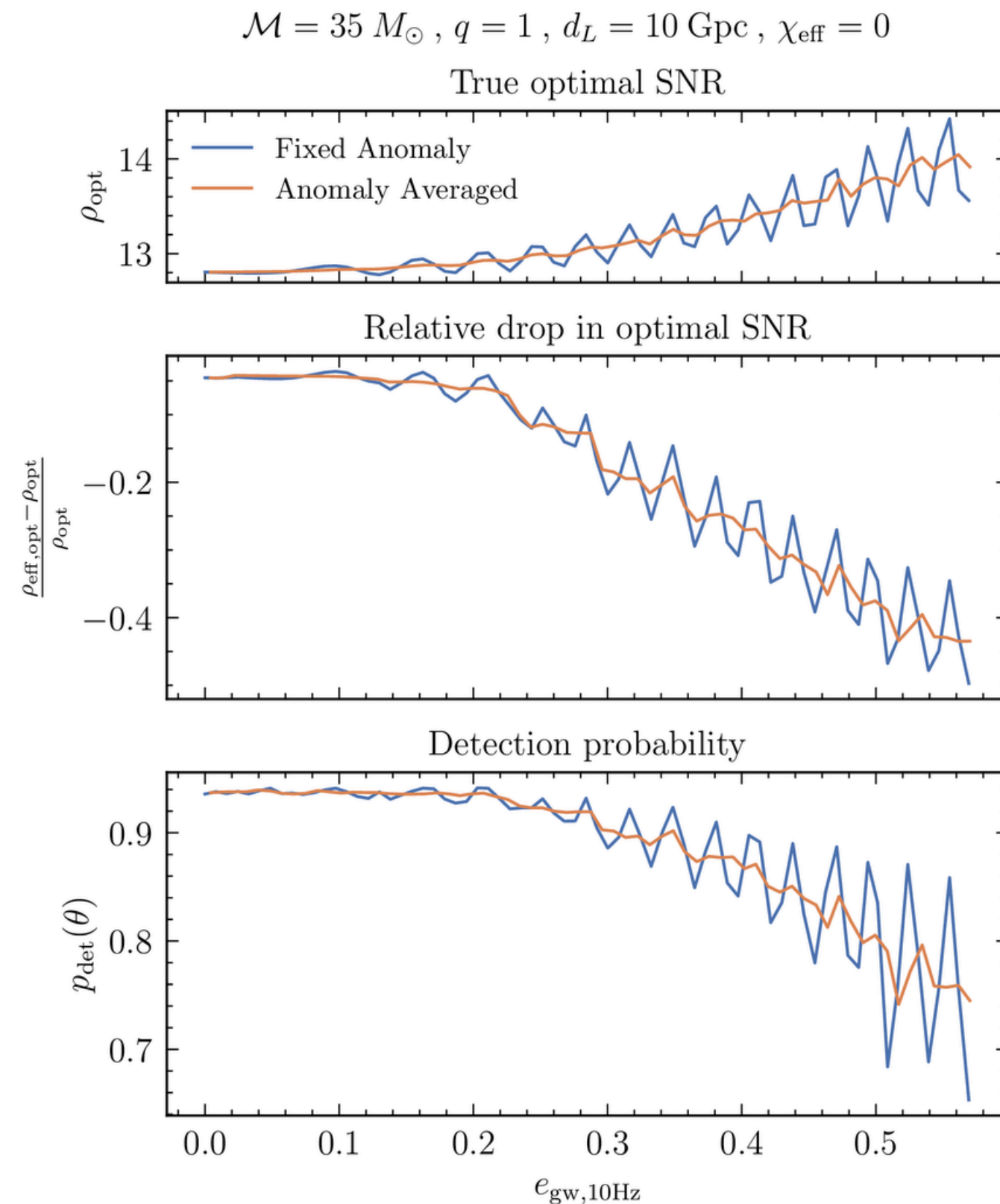
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Why the oscillations?

Less energy emission, low e



More energy emission, medium e



Most energy emission, high e



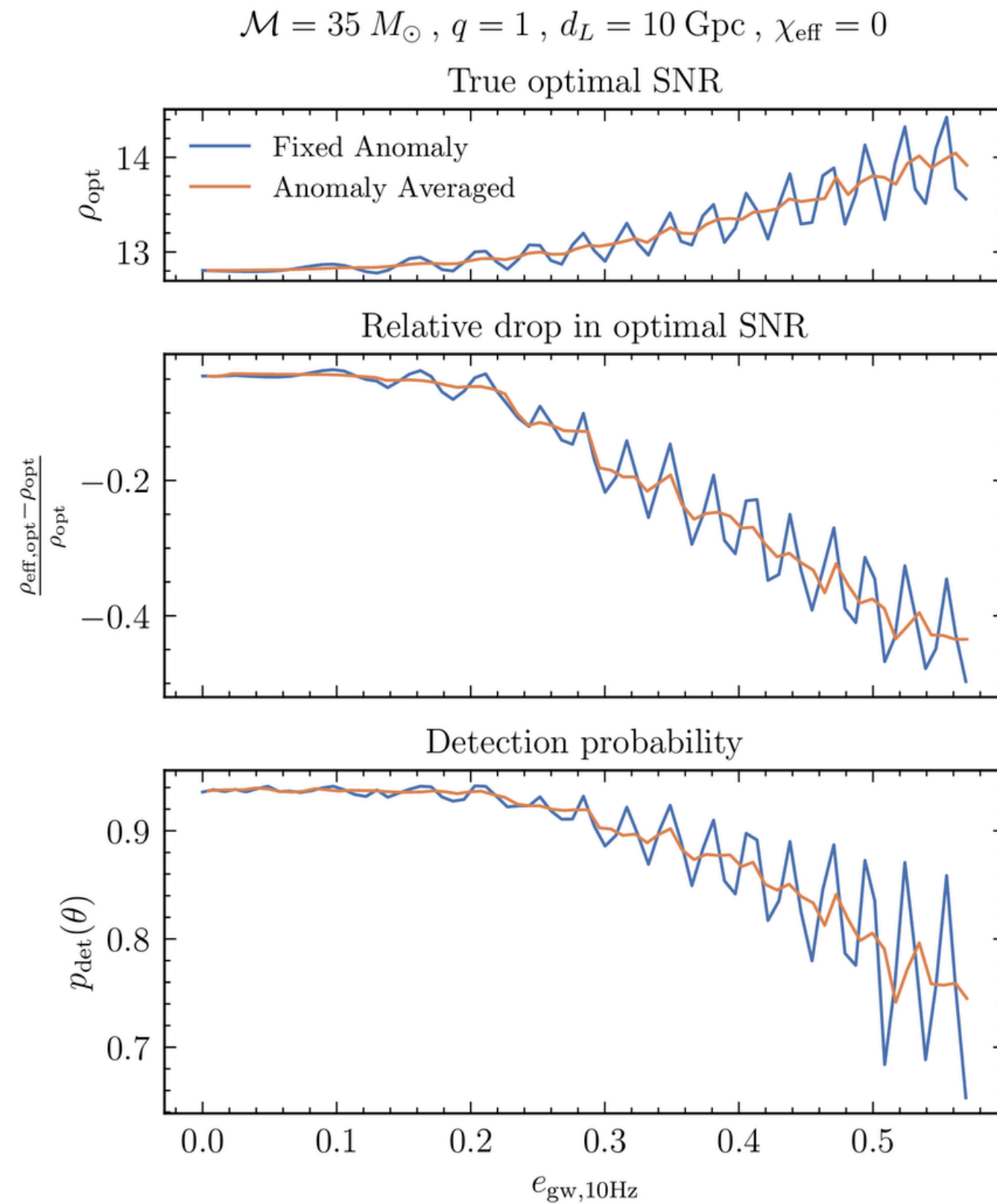
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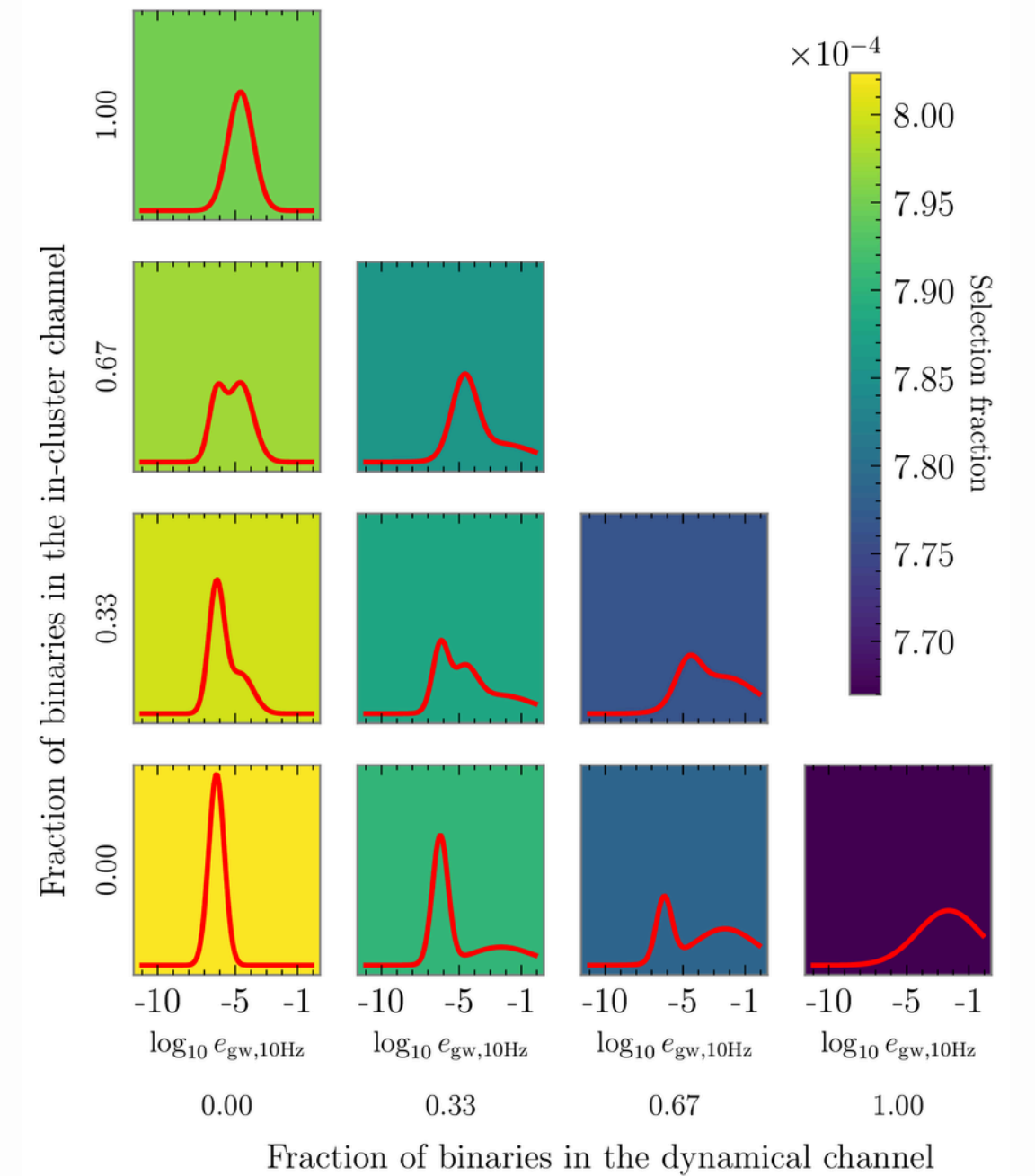
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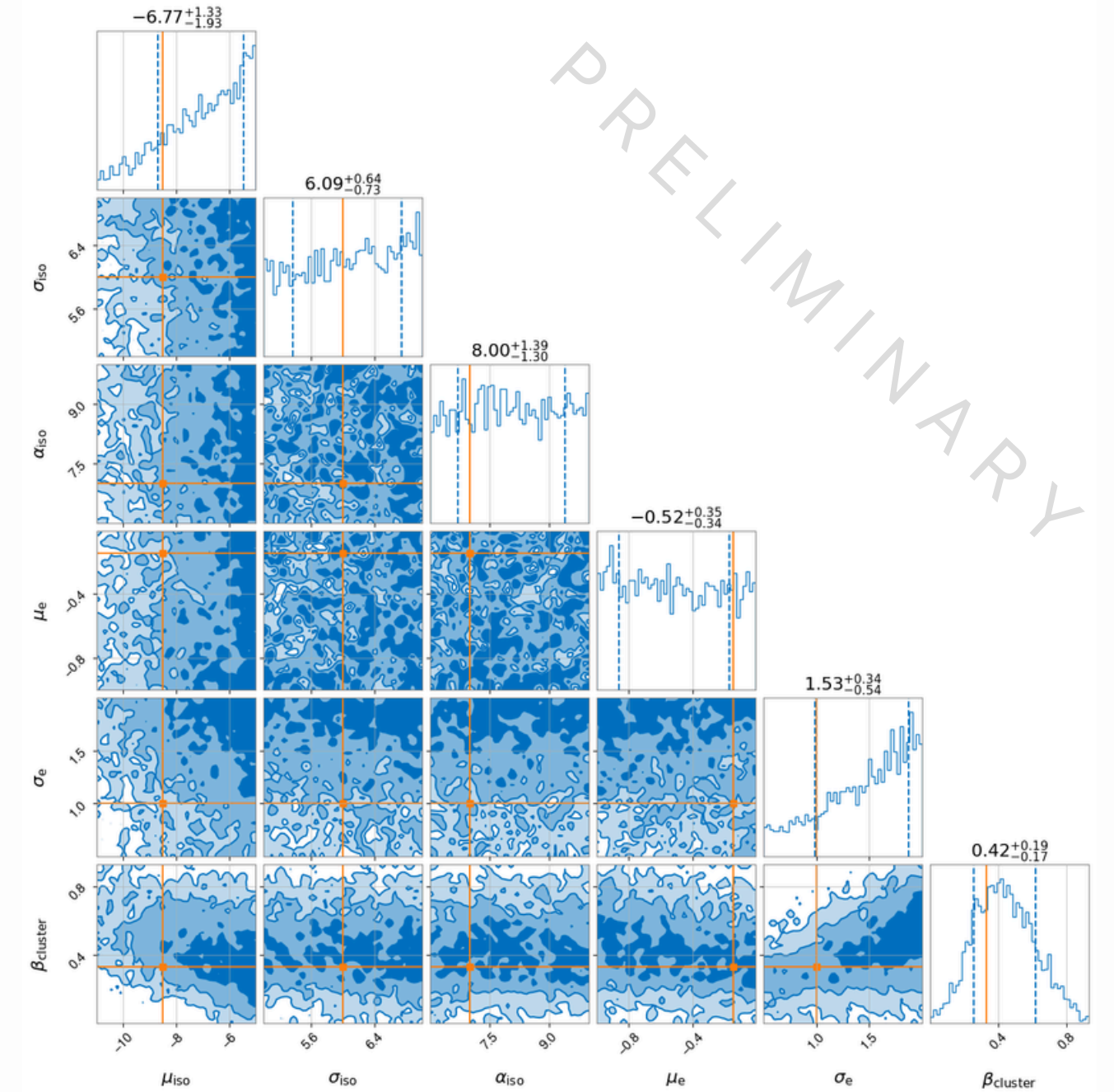
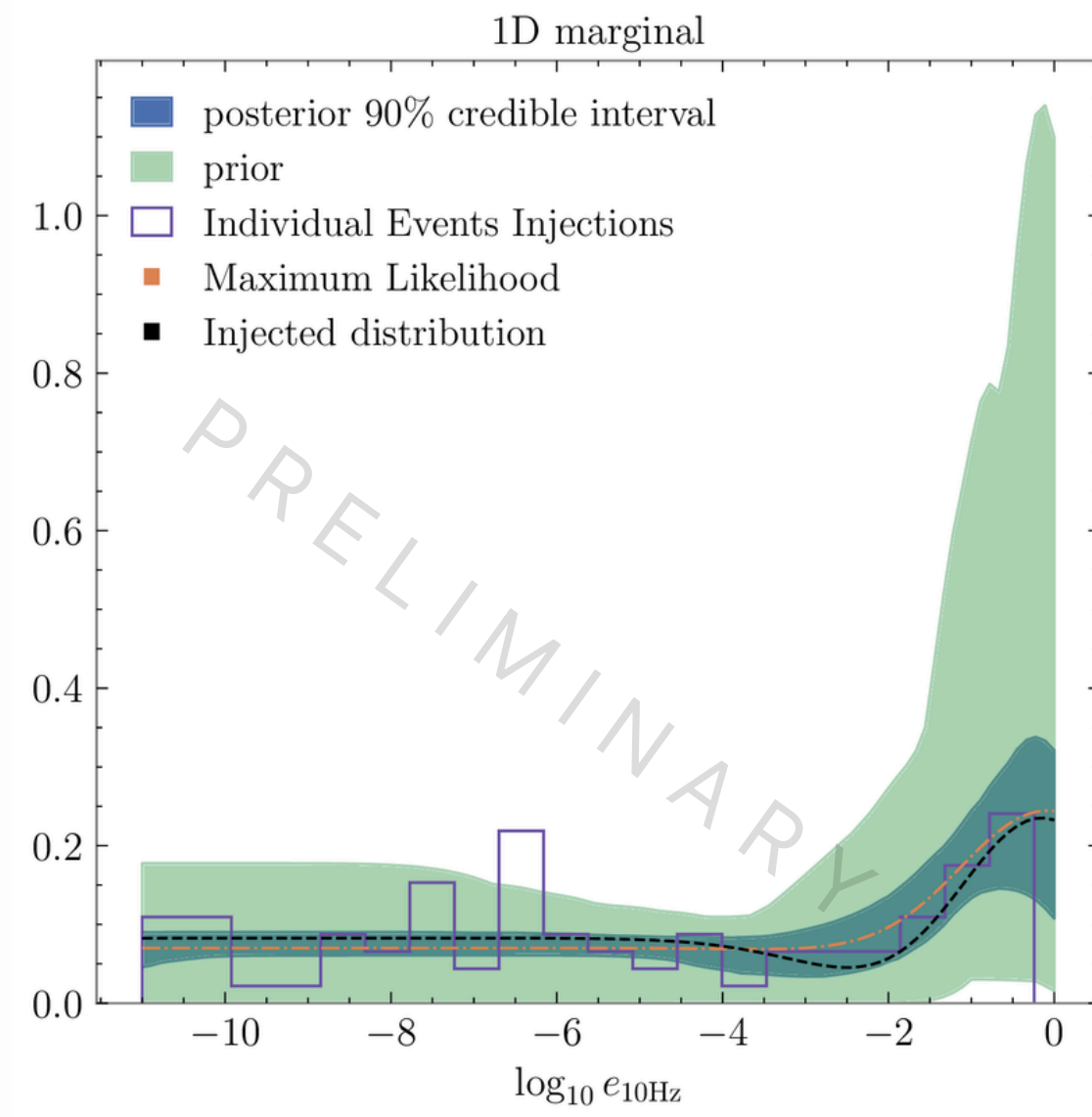


04 We show different realizations of the gaussian mixture population and how detectable it is

POPULATION INJECTION AND RECOVERY

We generate an injection with 33% of binaries coming from the eccentric channel and 67% from the non-eccentric channel. We inject 100 events total.

We use GWPopulation (Talbot+ 2019, 2025) to do our recovery and DINGO-IS (Dax+ 2021, 2022) to do individual event inference.



Gupte+2025 (in prep) See also
Zeeshan+ 2024, Zevin+ 2021

IN CONCLUSION

- Eccentricity is an extremely valuable parameter for GW inference and it's only a matter of time before confidently observed.
- By combining events hierarchically, we can learn about different formation channels

LOTS OF TODOS!

Real data

We want to do hierarchical inference on real GW events from O3.

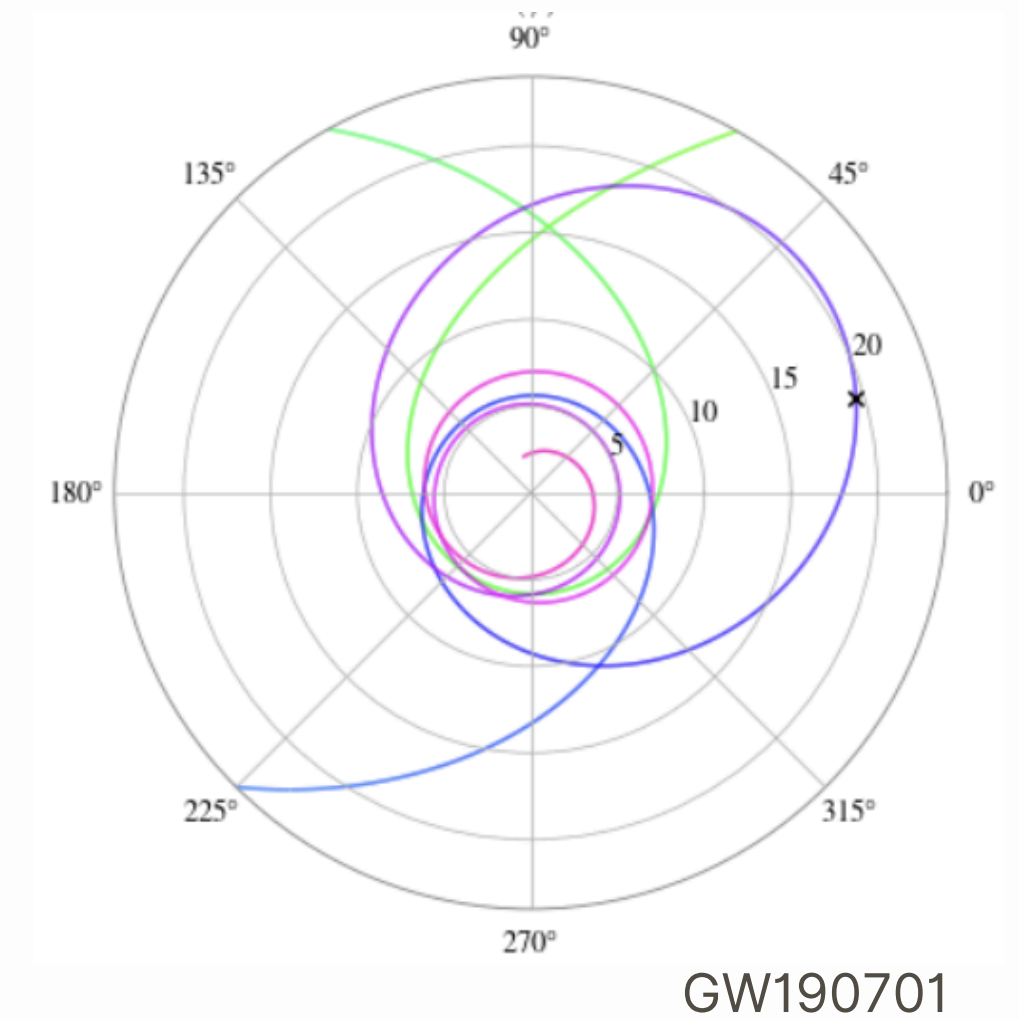
With recent updates to DINGO (Dax+2024) we can also now handle lower mass signals

Flexible Pop. Models

We want to use a flexible model population model to do transdimensional MCMC on the population.

Spin & Mass Correlations

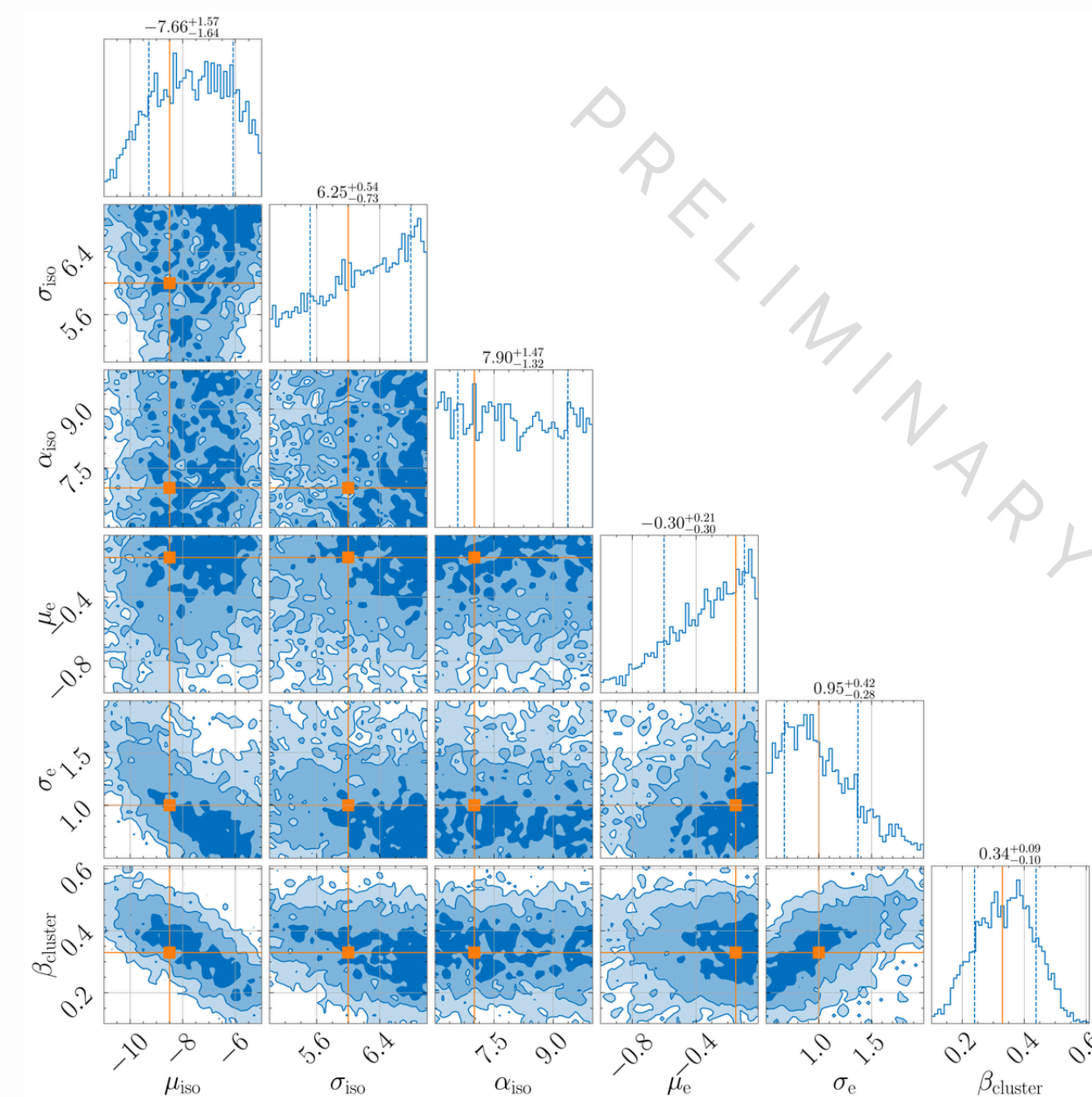
We would like to correlate eccentricity with spins and masses to see if we can obtain better constraints



APPENDIX

POPULATION INJECTION AND RECOVERY

(ZERO NOISE)

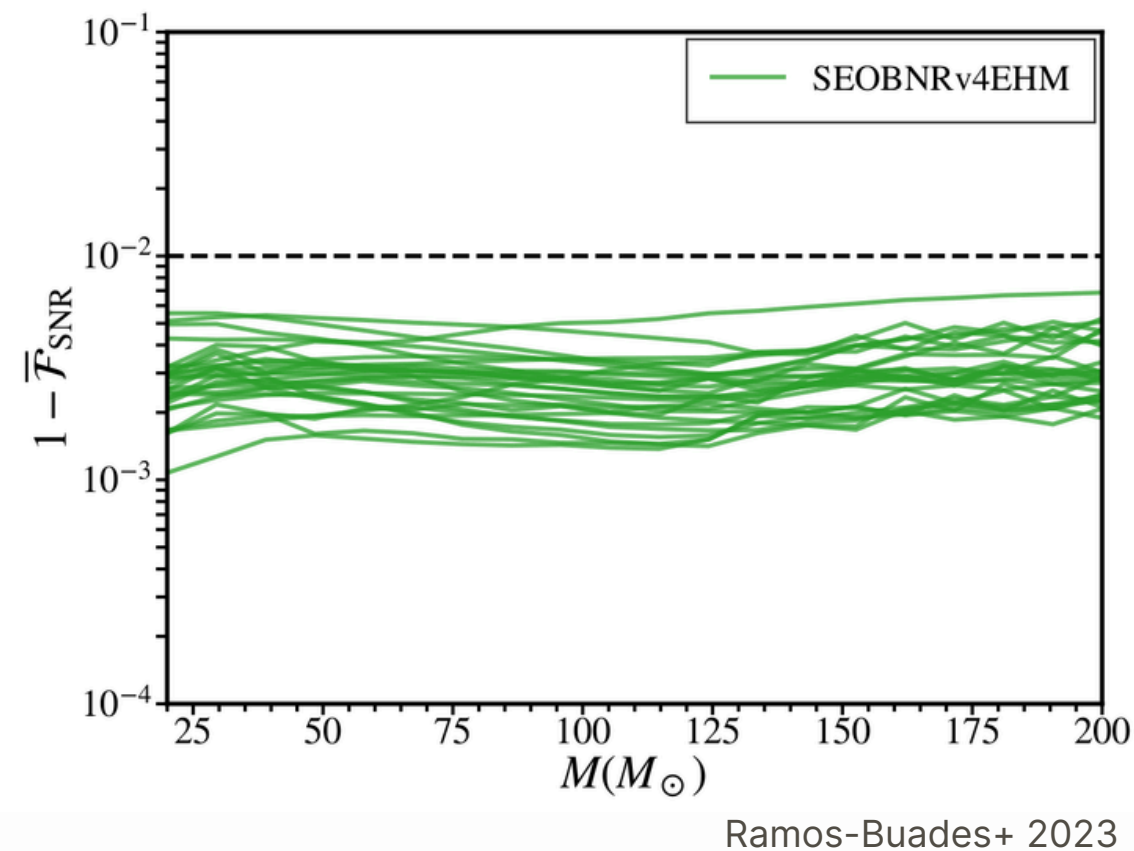
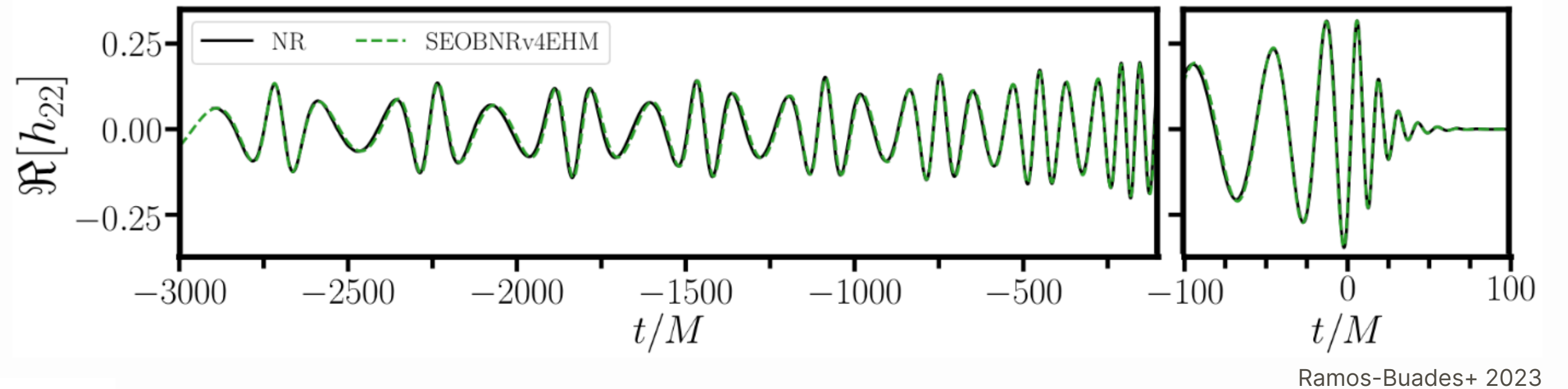


Gupte+2025 (in prep) See also
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WHICH WAVEFORM MODEL?

Fortunately, in recent years there have been a plethora of eccentric waveform models: Huerta+2018, Hinder+2018, Cao+2017, Setyawati+2021, Nagar+2022, Islam+2021, Gamboa+2024, Planas+ 2025, Morras+2025

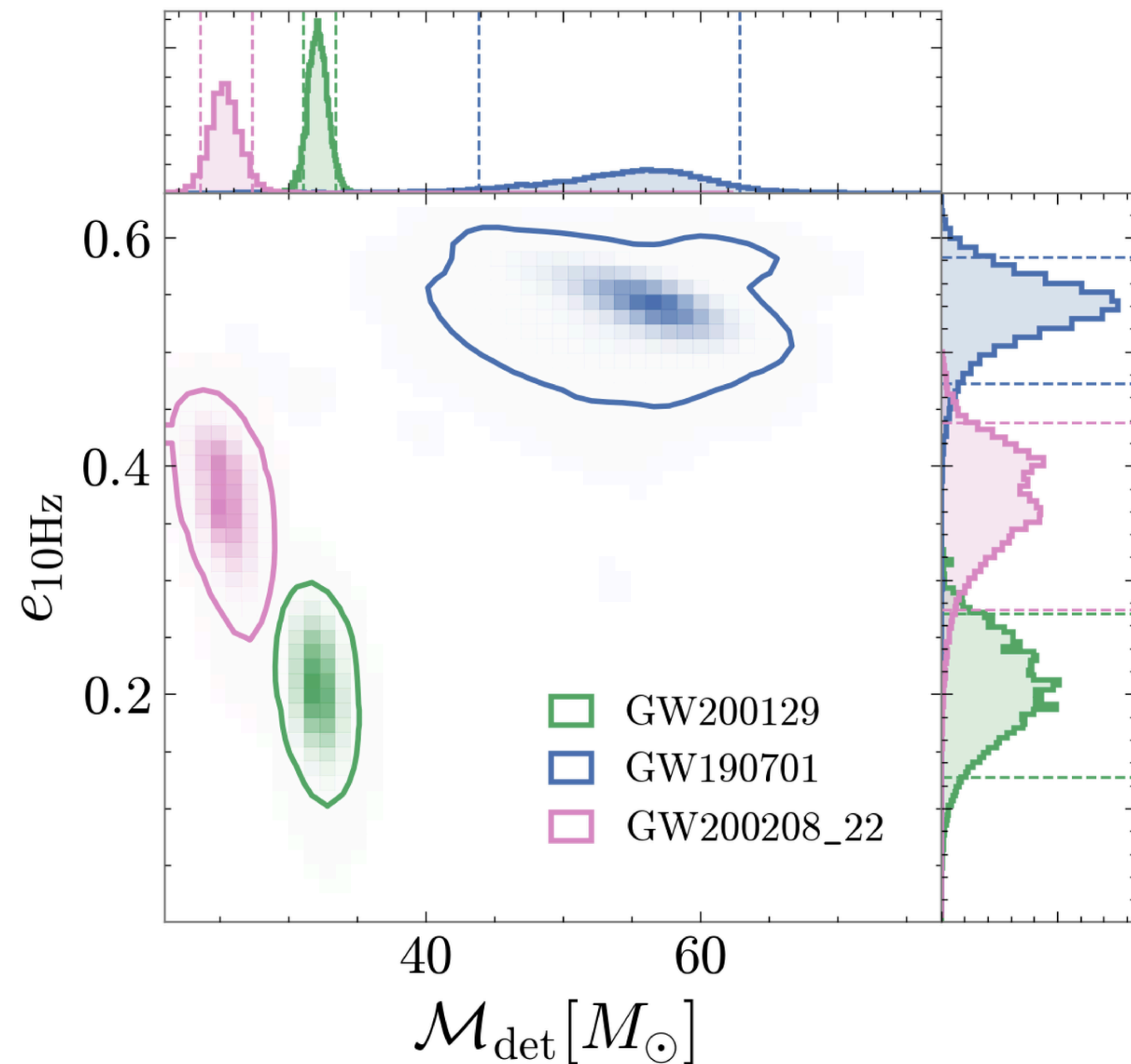
We use the **SEOBNRv4EHM** waveform model from Ramos-Buades+ 2023. Later we will use the upgraded version of this model **SEOBNRv5EHM** from Gamboa+ 2024.



- Inspiral merger ringdown waveform
- Models higher order modes
- Models both relativistic anomaly and the eccentricity
- Validated against NR until $e_{20\text{hz}} < 0.3$
- Parameter estimation runs take $O(4-5 \text{ days})$ with nested sampling

POST PROCESSING THE OUTPUT OF ASTROPHYSICS SIMULATIONS

We analyze the events with SEOBNRv5EHM (Gamboa+ 2024). We find no longer see railing in GW190701



We post-process the output of the CMC simulation to obtain eccentricity definitions consistent with GW data analysis (Vijaykumar+ 2024, Shaikh+ 2023)

